

(Purely) Synthetic Block Copolymers *- Molecular Characterization -*

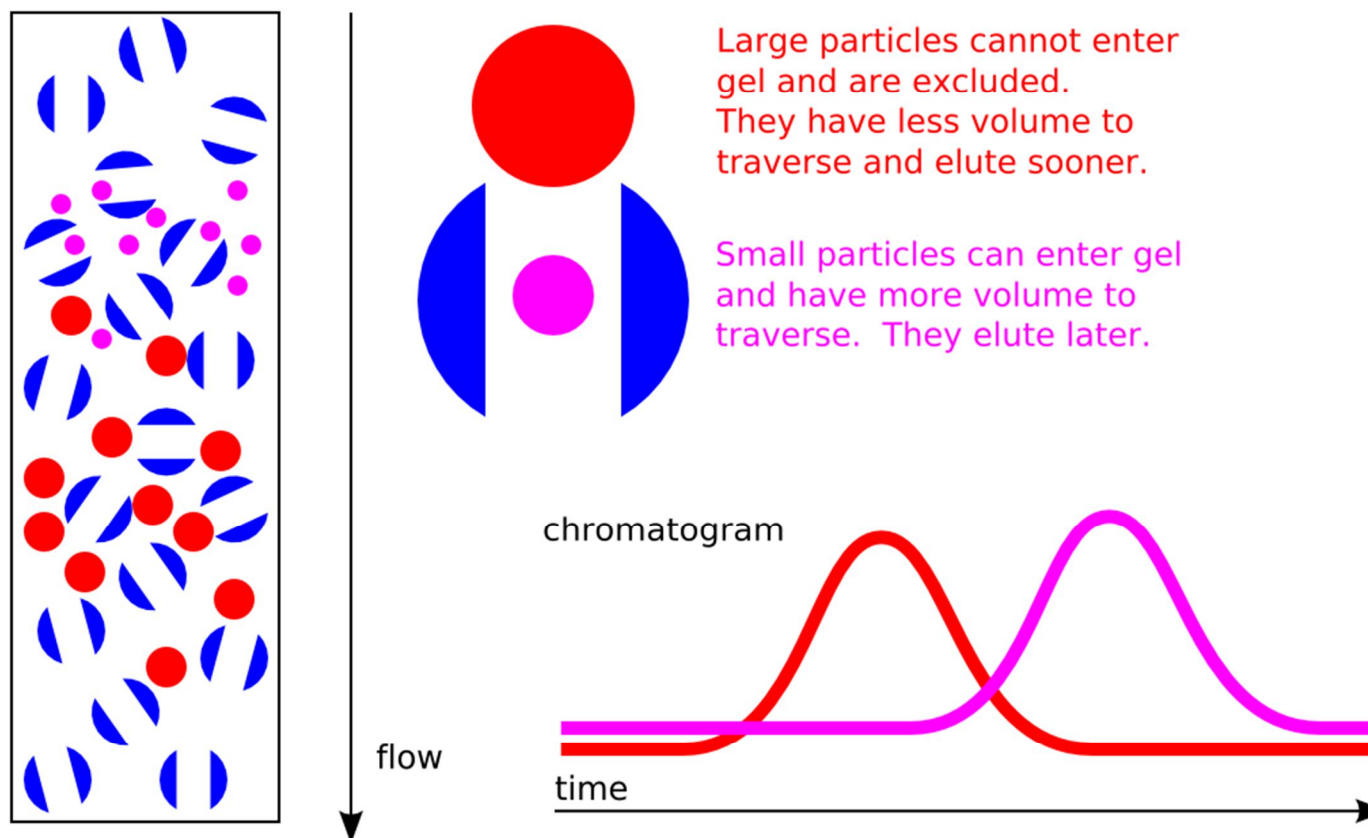
Questions we need to answer:

- **How do I prove I made a block copolymer and not just ended up with a mixture of homopolymers?**
- **What are the fractions of my respective blocks, i.e., how long is my second block?**
- **How homogeneous in size are my chains?**

Molecular Characterization of Block Copolymers

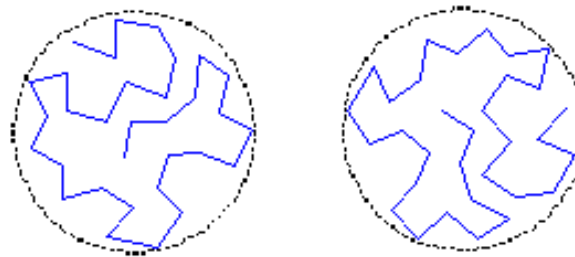
Which methods can we use?

Size-Exclusion Chromatography



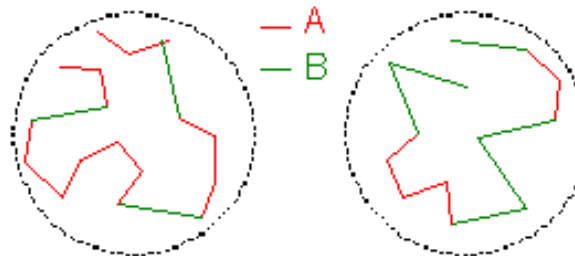
Size-Exclusion Chromatography

A) **Homo polymer** $P_n = 31$; $V_h = f(M)$



same V_h thus same M

B) **Copolymers** with same V_h

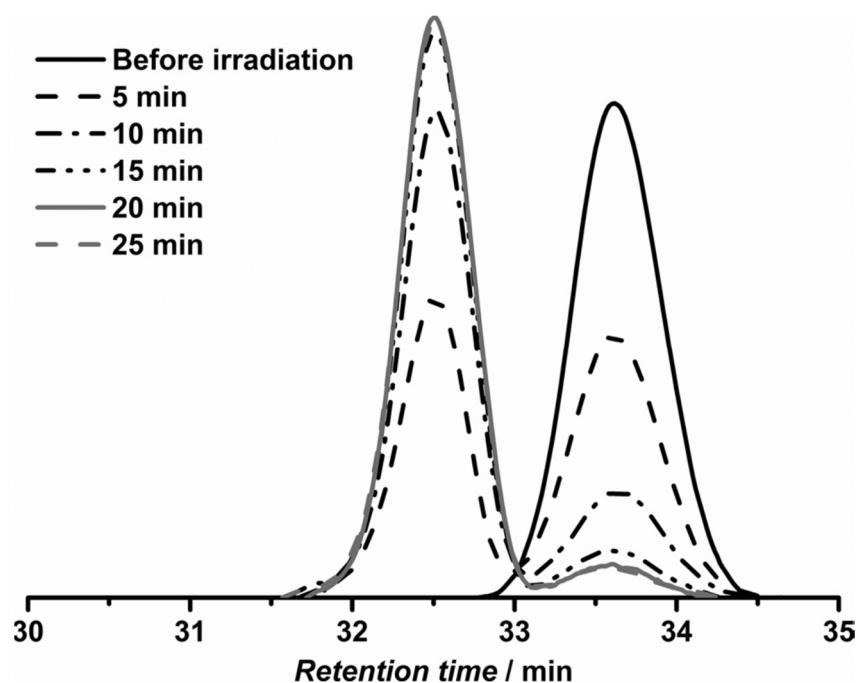
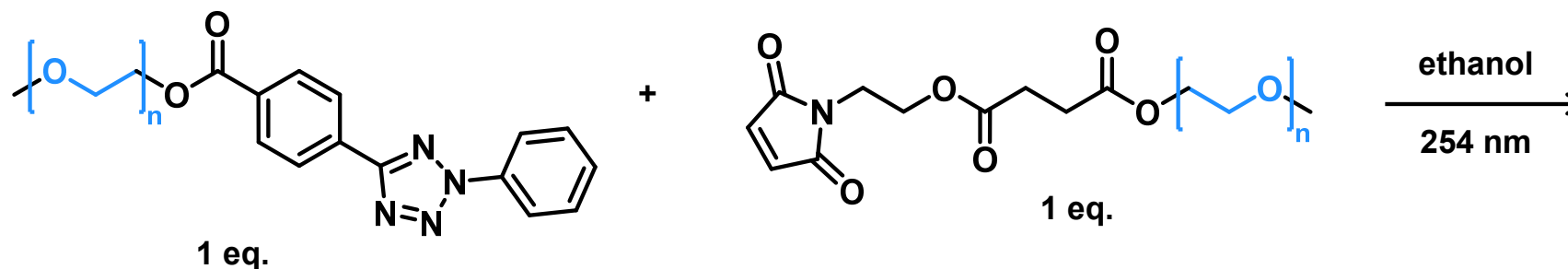


$P_n = 16$; $x_A = 0.81$ $P_n = 12$; $x_A = 0.46$

different composition and chain length

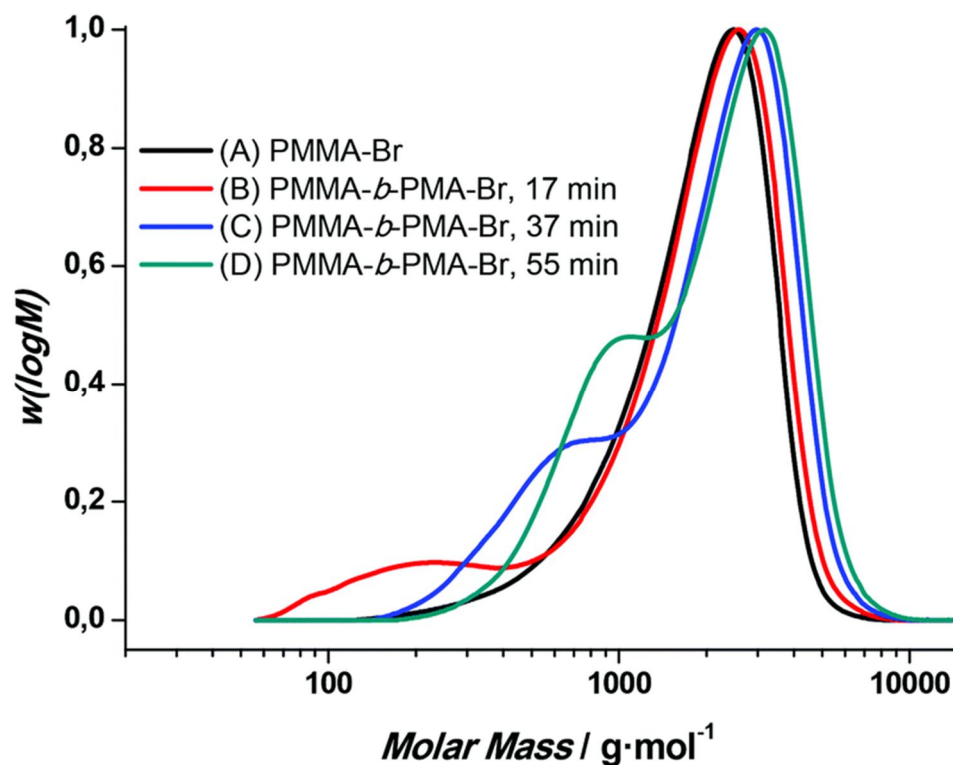
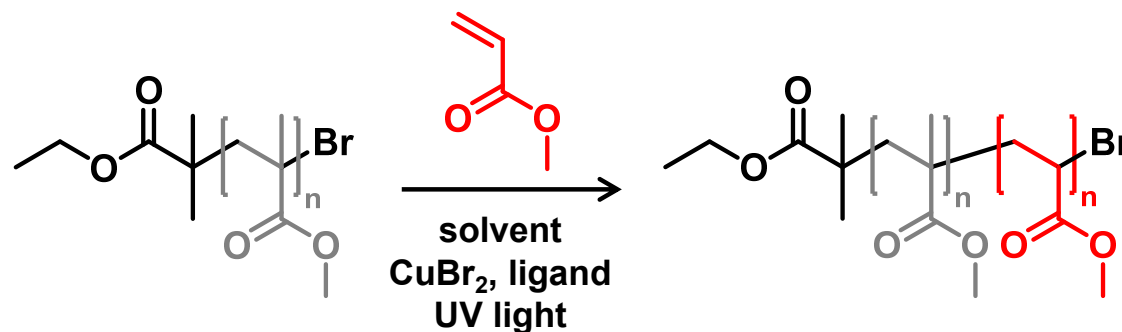
Molecular Characterization of Block Copolymers

The “easy” case: coupling of two polymer strands



Molecular Characterization of Block Copolymers

The classic case: polymerization from a macromolecular reactant



Polym. Chem. 2015, 6, 6488

Molecular Characterization of Block Copolymers

The classic case: polymerization from a macromolecular reactant

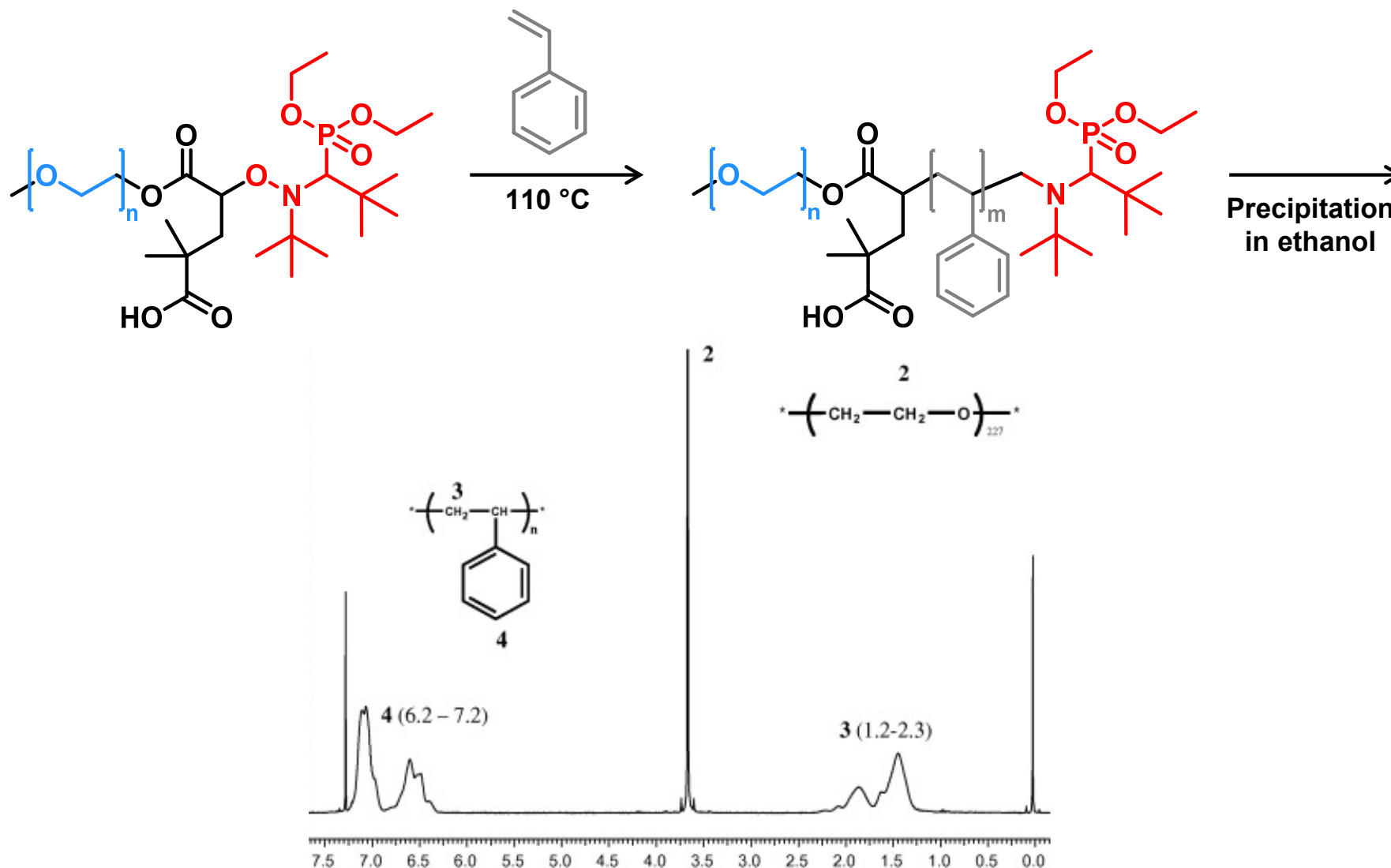
Similar problems may happen with, for instance:

- **NMP of styrene from a macroalkoxyamine**
- **ROP of lactones from a hydroxyl-terminated polymer**

WHY?

Molecular Characterization of Block Copolymers

Nuclear Magnetic Resonance | Simple Experiments (^1H , ^{13}C , ^{19}F , etc)

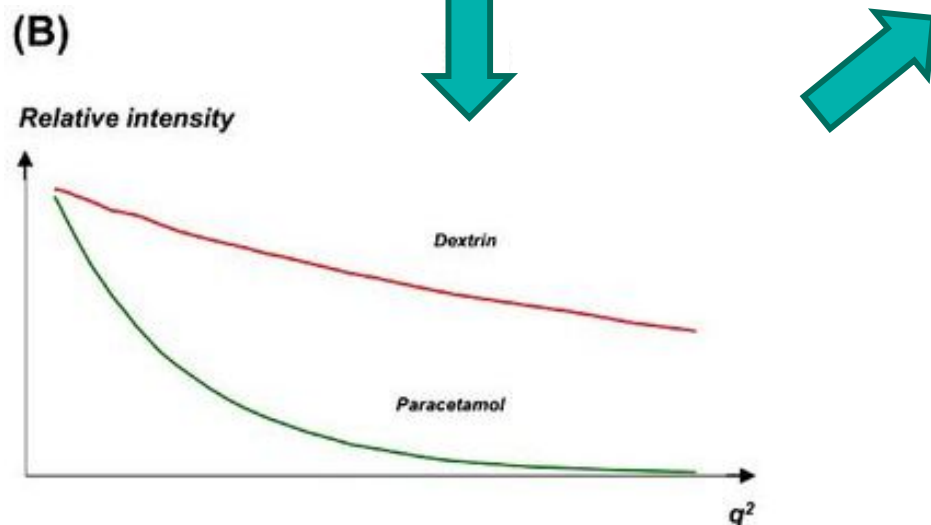
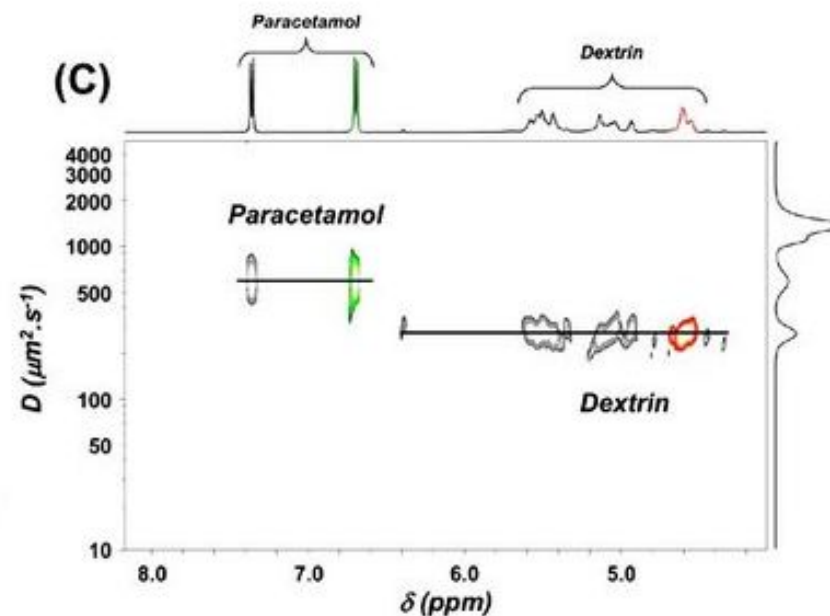
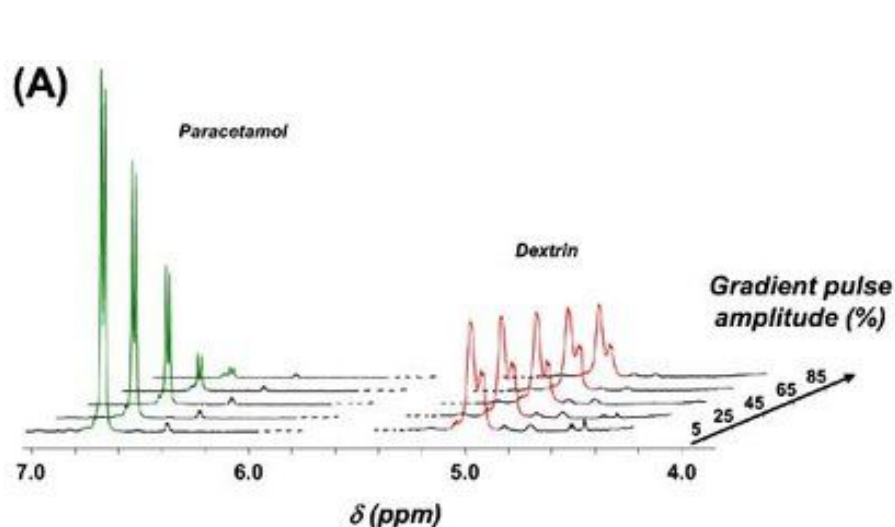


But only a real proof after full precipitation in selective solvent!

Micropor. Mesopor. Mater. **2008**, 112, 612

Molecular Characterization of Block Copolymers

Nuclear Magnetic Resonance | Diffusion-Ordered Spectroscopy (DOSY)



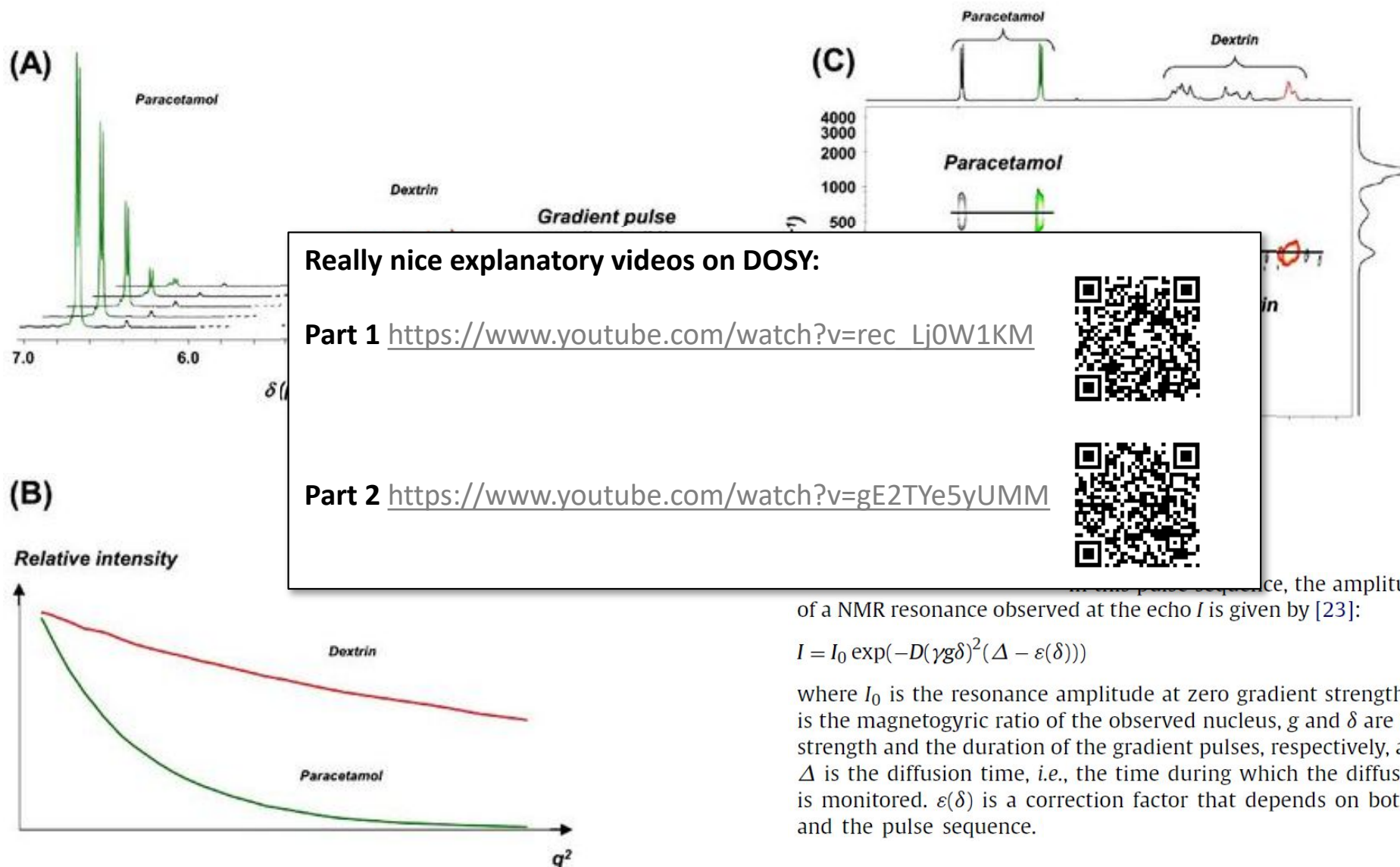
In this pulse sequence, the amplitude of a NMR resonance observed at the echo I is given by [23]:

$$I = I_0 \exp(-D(\gamma g \delta)^2 (\Delta - \varepsilon(\delta)))$$

where I_0 is the resonance amplitude at zero gradient strength, γ is the magnetogyric ratio of the observed nucleus, g and δ are the strength and the duration of the gradient pulses, respectively, and Δ is the diffusion time, i.e., the time during which the diffusion is monitored. $\varepsilon(\delta)$ is a correction factor that depends on both δ and the pulse sequence.

Molecular Characterization of Block Copolymers

Nuclear Magnetic Resonance | Diffusion-Ordered Spectroscopy (DOSY)



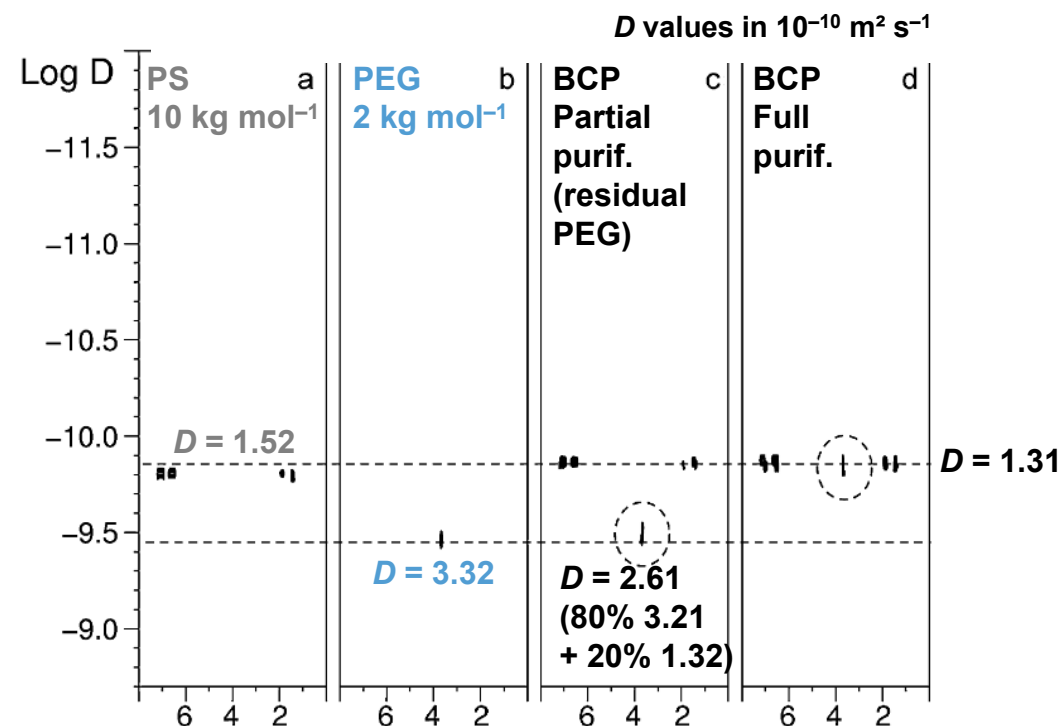
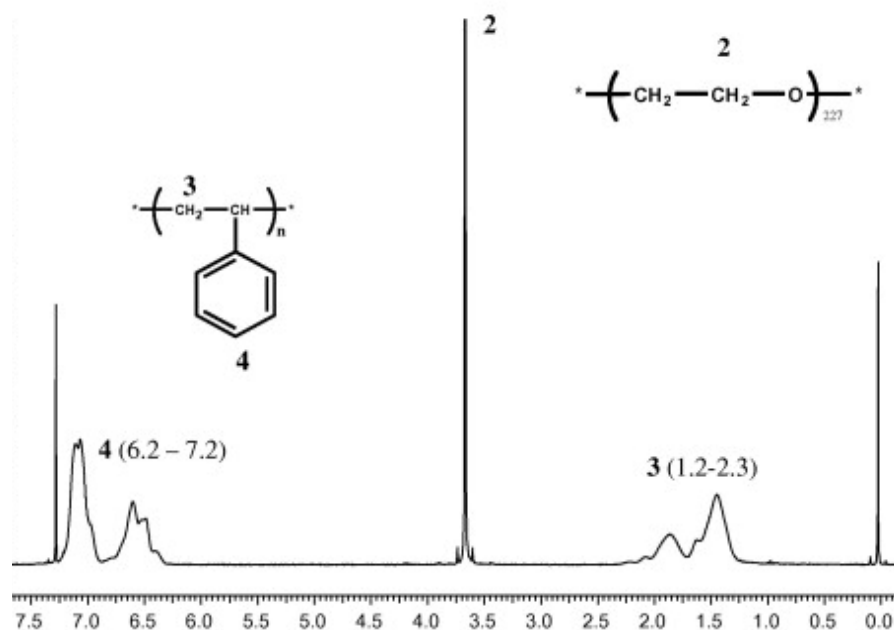
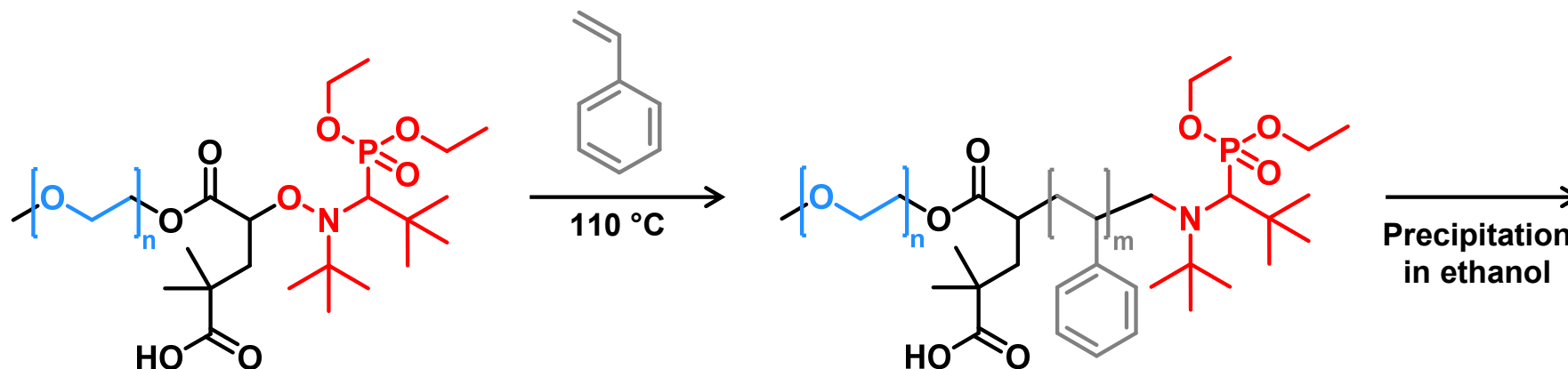
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Molecular Characterization of Block Copolymers

Nuclear Magnetic Resonance | Diffusion-Ordered Spectroscopy (DOSY)



Anal. Chem. Acta **2009**, 654, 45

*The main idea is to attribute **chemical information** to a **size distribution**.*

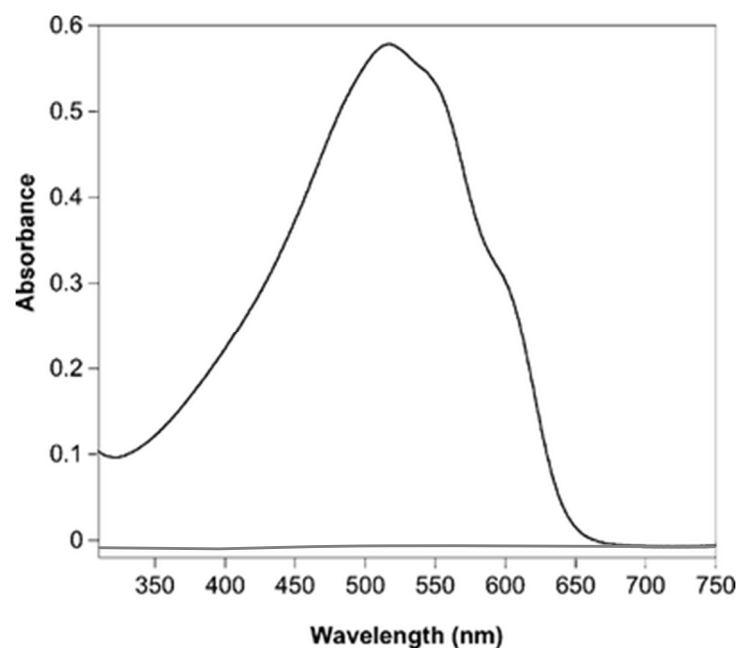
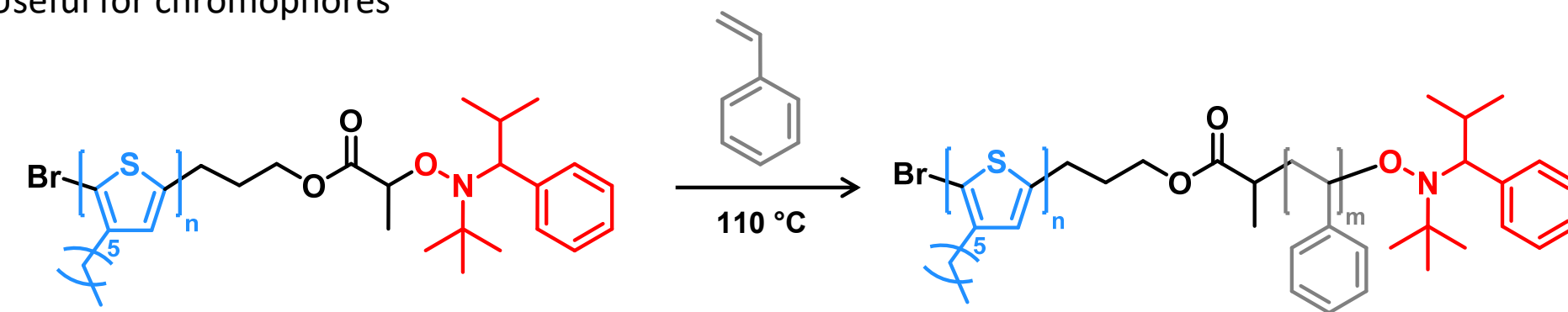


What can then be coupled to SEC?

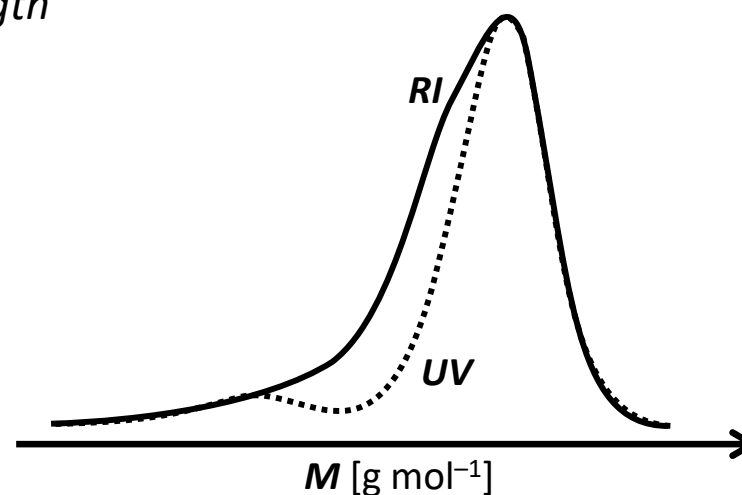
Coupling SEC to Other Characterization Methods ("Hyphenation")

UV Detection

Useful for chromophores



Fixed wavelength
Ex.: 450 nm



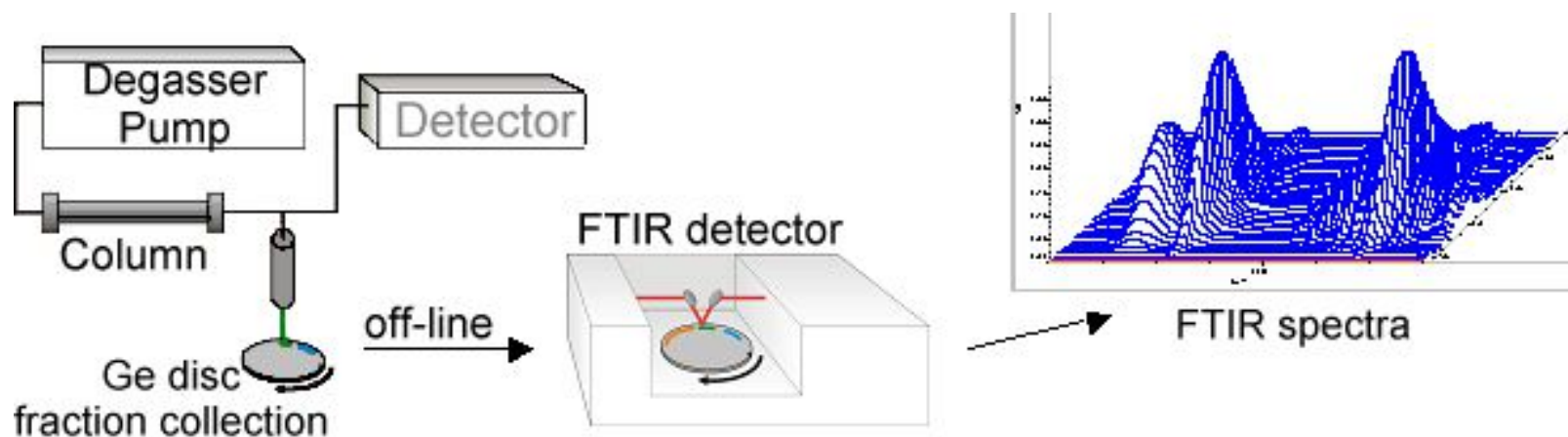
Polym. Chem. 2012, 3, 1693

Molecular Characterization of Block Copolymers

Coupling SEC to Other Characterization Methods

FT-IR Spectroscopy

Offline (e.g., sold by PSS - Mainz) – not hyphenated *per se*



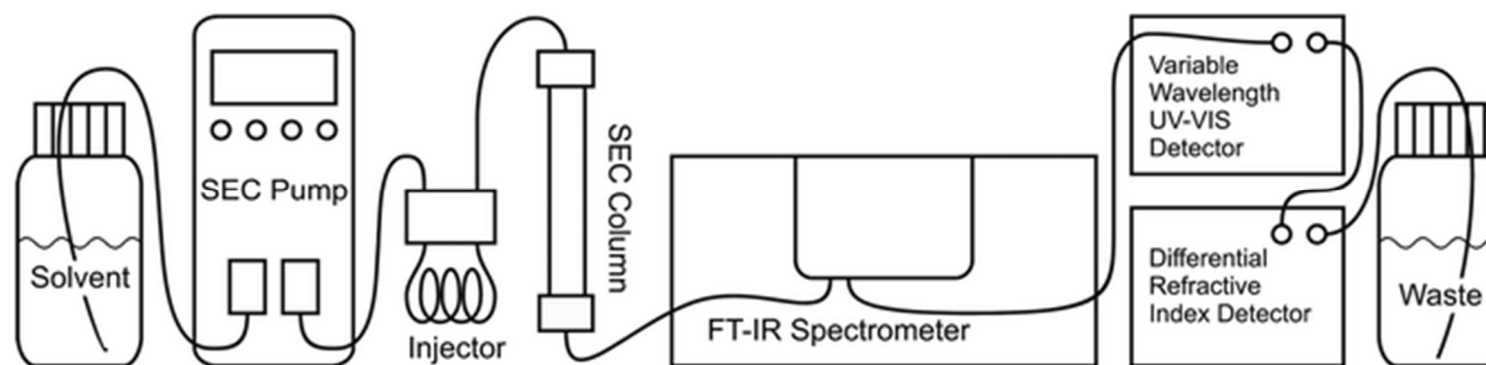
J. Chromatogr. A **2003**, 1017, 83

Coupling SEC to Other Characterization Methods

FT-IR Spectroscopy

Online

(Prof. M. Wilhelm,
Karlsruhe)



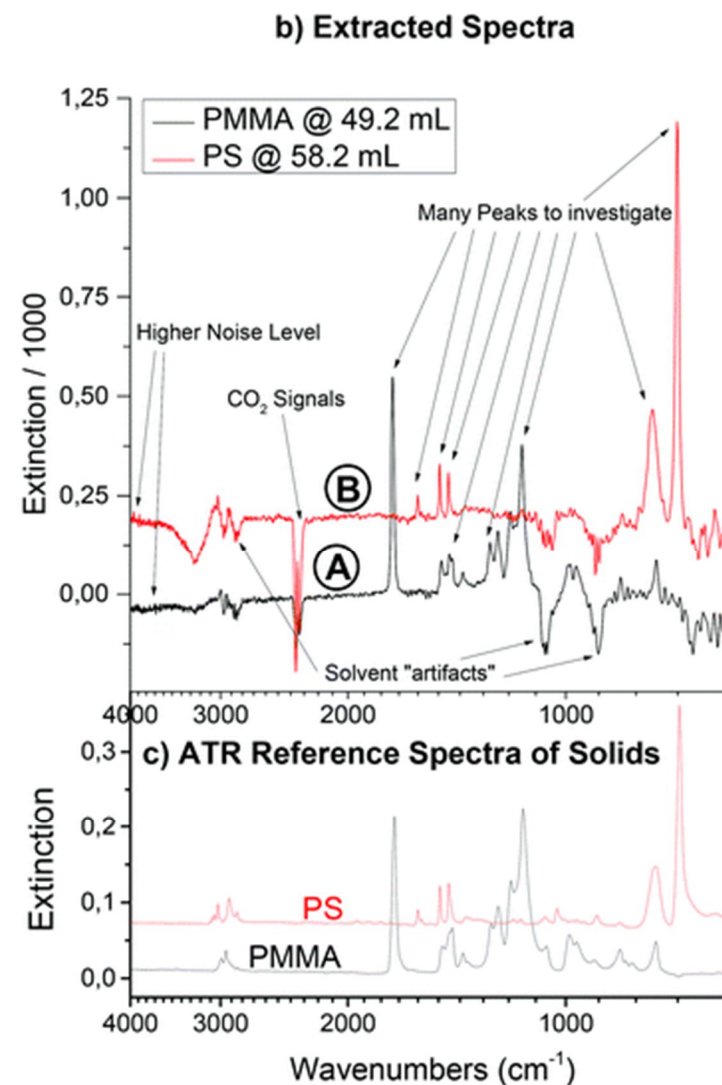
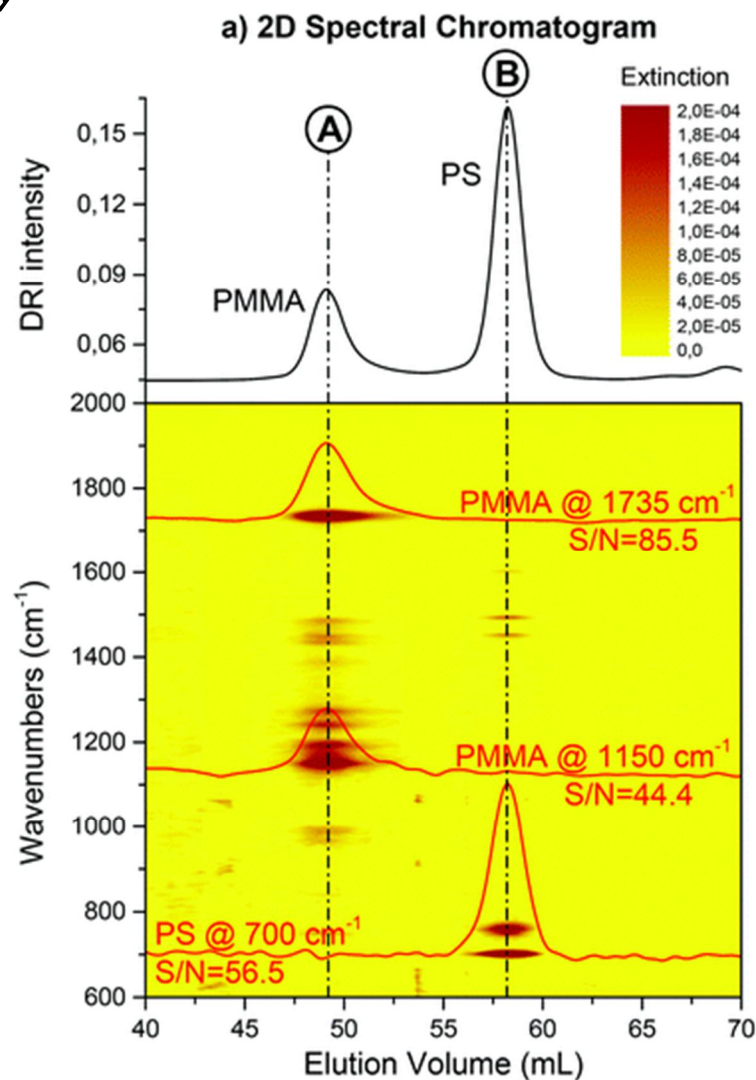
Molecular Characterization of Block Copolymers

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FT-IR Spectroscopy

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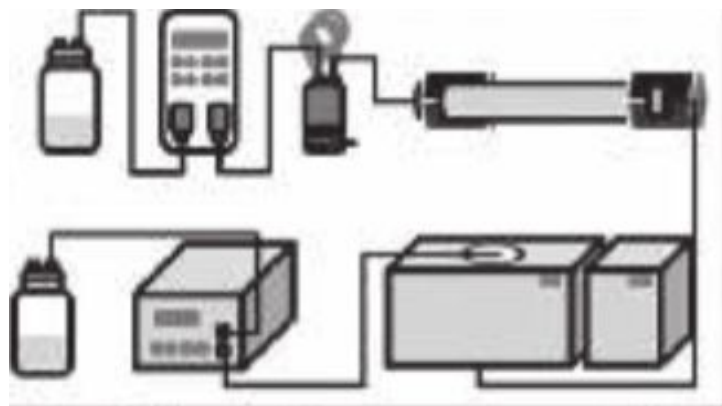
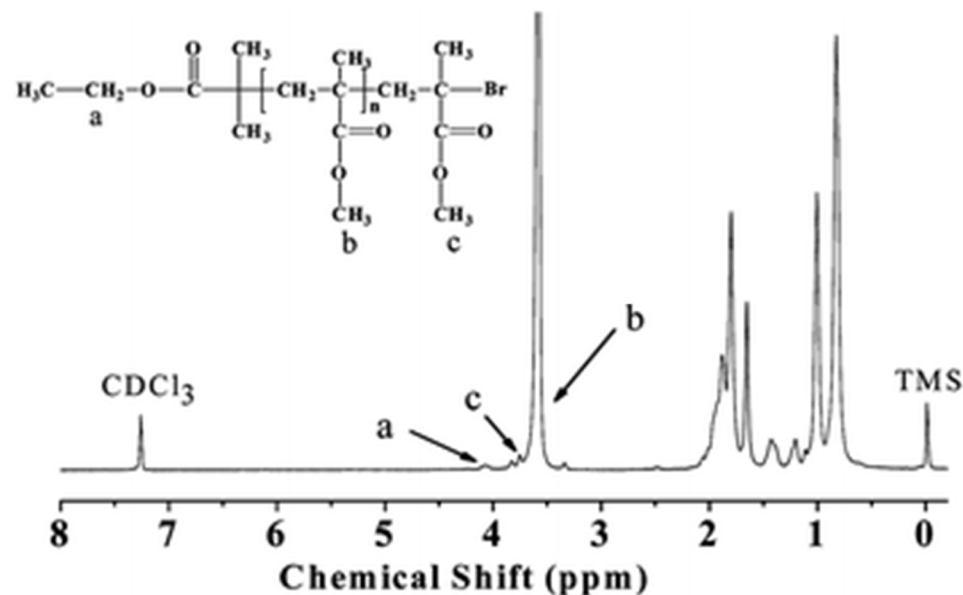


Polym. Chem. **2015**, 6, 128

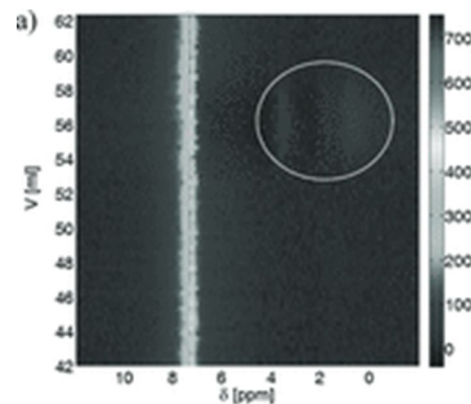
Molecular Characterization of Block Copolymers

Coupling SEC to Other Characterization Methods

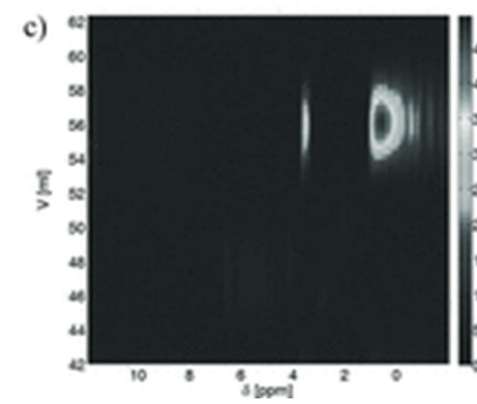
NMR Spectroscopy (AK Wilhelm - KIT)



Raw Data



Solvent subtraction
and more

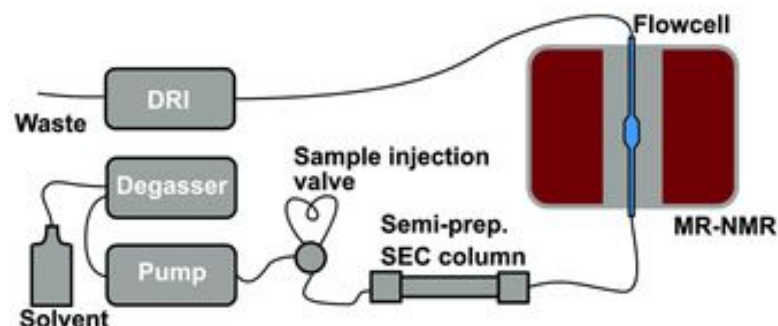


Macromol. Chem. Phys. **2012**, 213, 1933

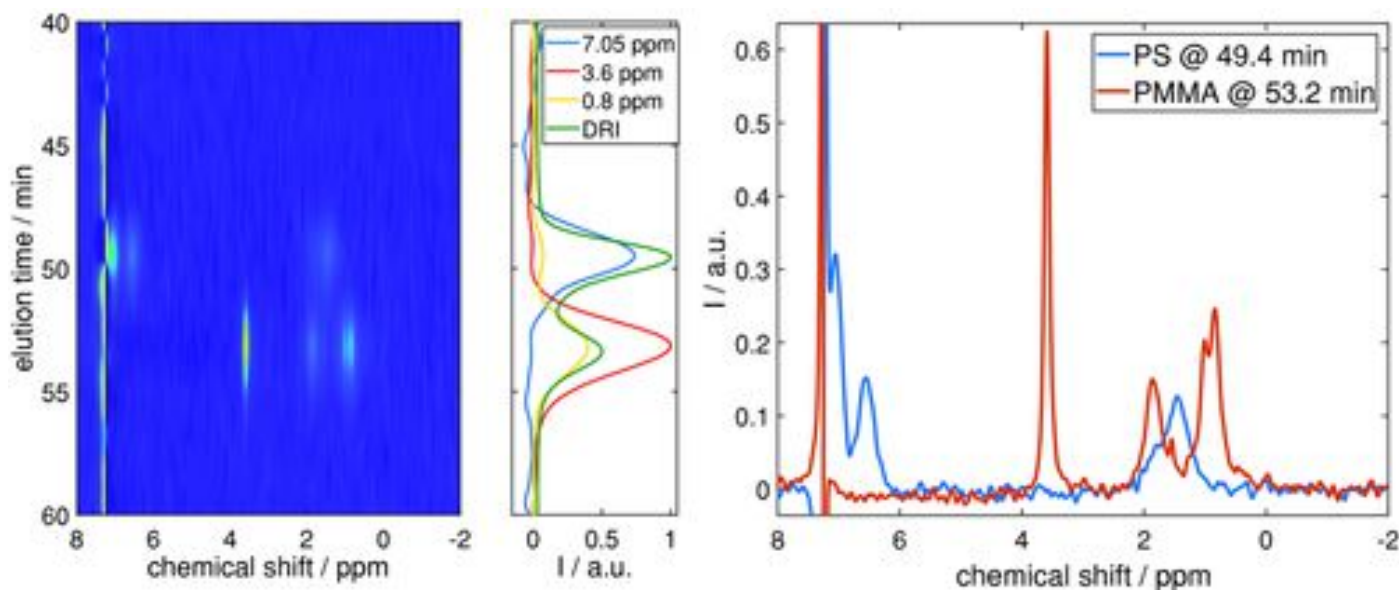
Molecular Characterization of Block Copolymers

Coupling SEC to Other Characterization Methods

NMR Spectroscopy (AK Wilhelm - KIT)



Deformulation of a blend of PMMA and PS

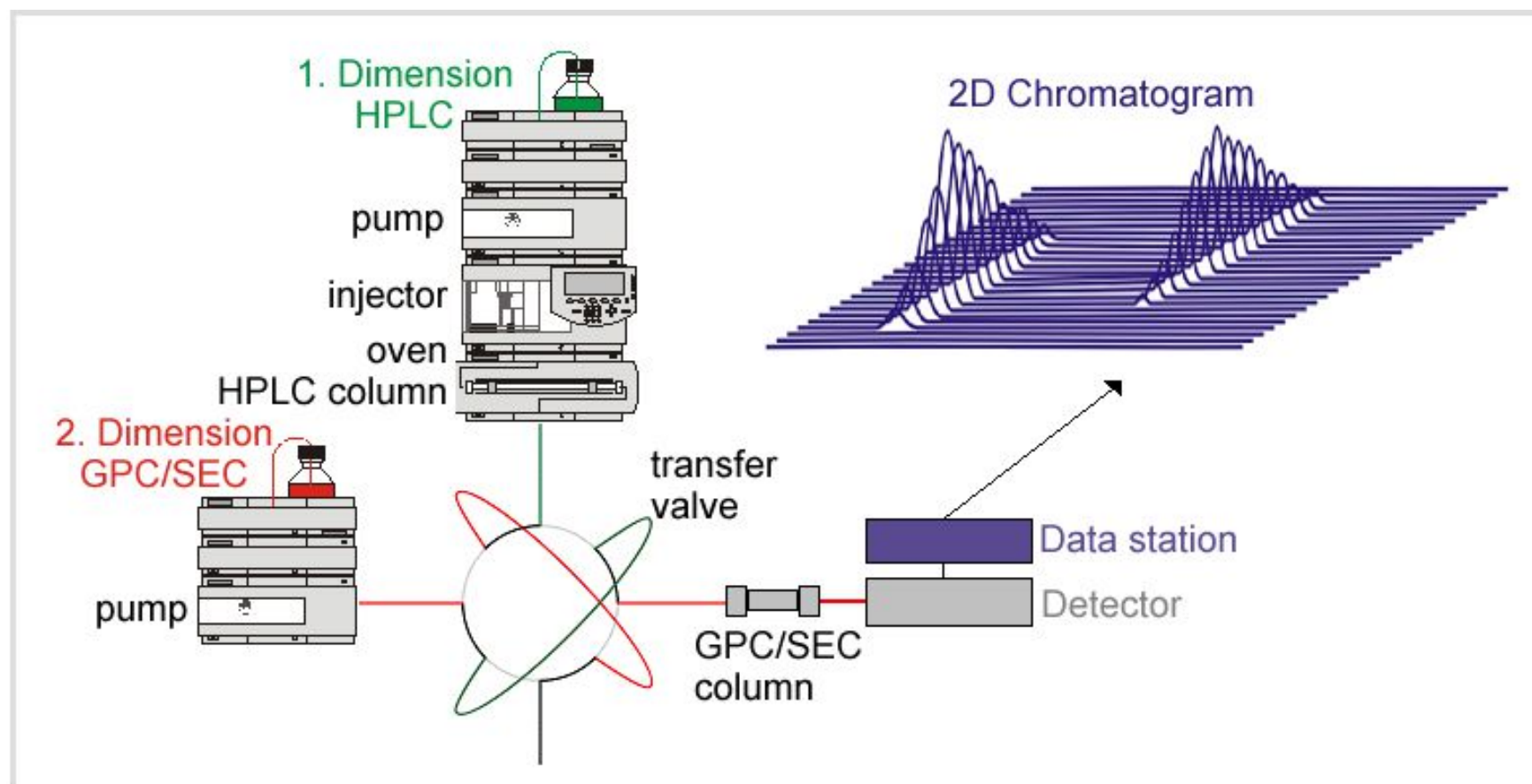


Macromol. Rapid Commun. **2018**, 39, 1700766

Molecular Characterization of Block Copolymers

Coupling SEC to Other Characterization Methods

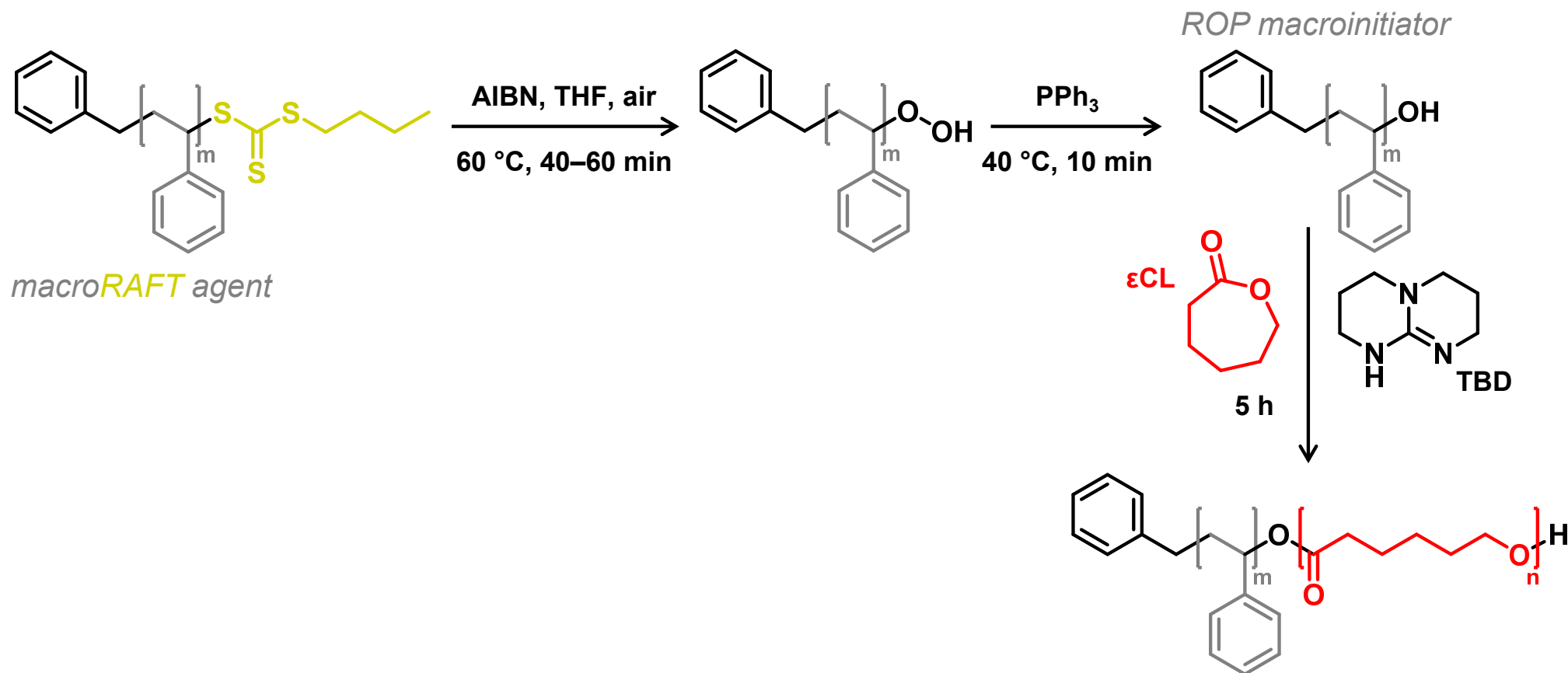
Two-Dimensional Size-Exclusion Chromatography



Molecular Characterization of Block Copolymers

Coupling SEC to Other Characterization Methods

Liquid Adsorption Chromatography at Critical Conditions–SEC (LACCC–SEC)

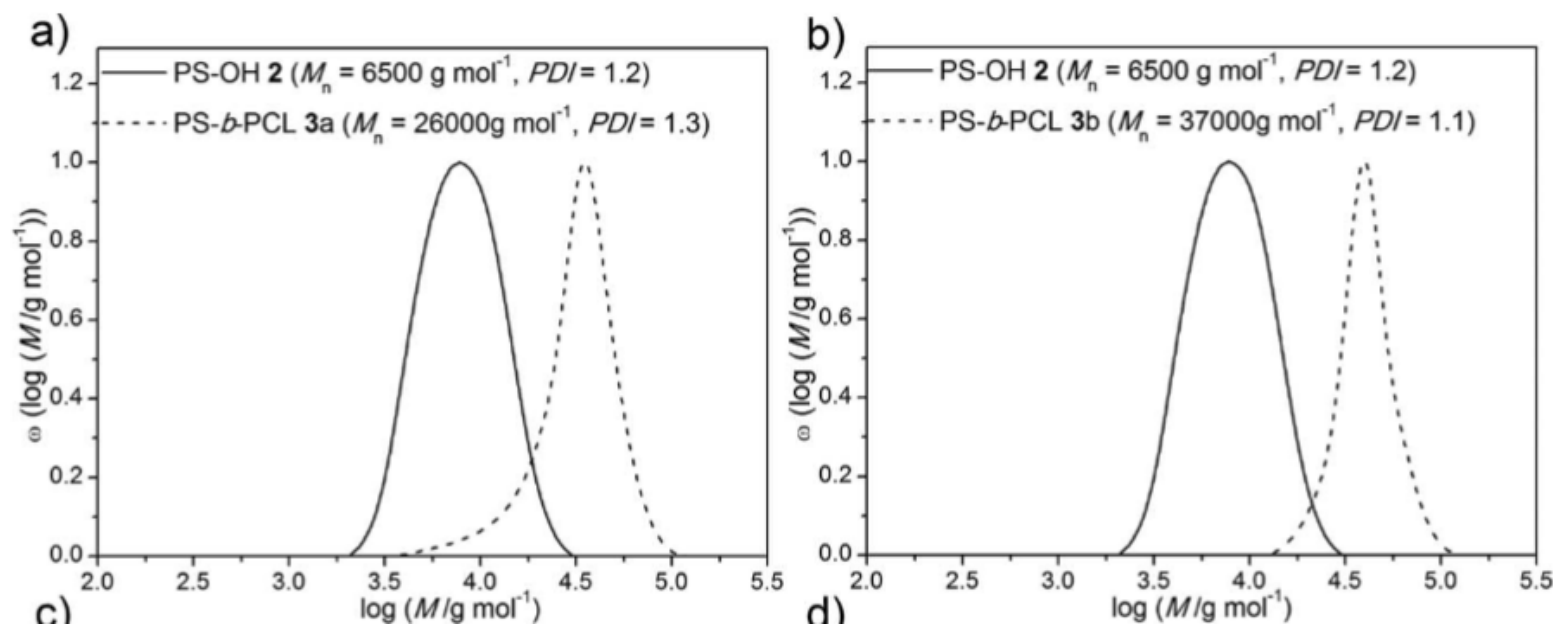


Polym. Chem. **2010**, 1, 634

J. Polym. Sci. A, Polym. Chem. **2011**, 49, 1

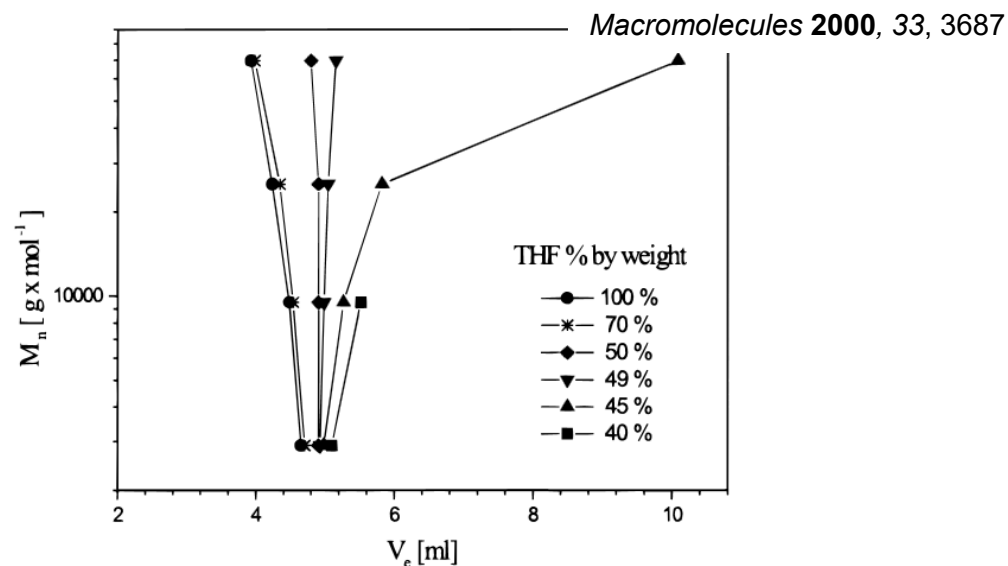
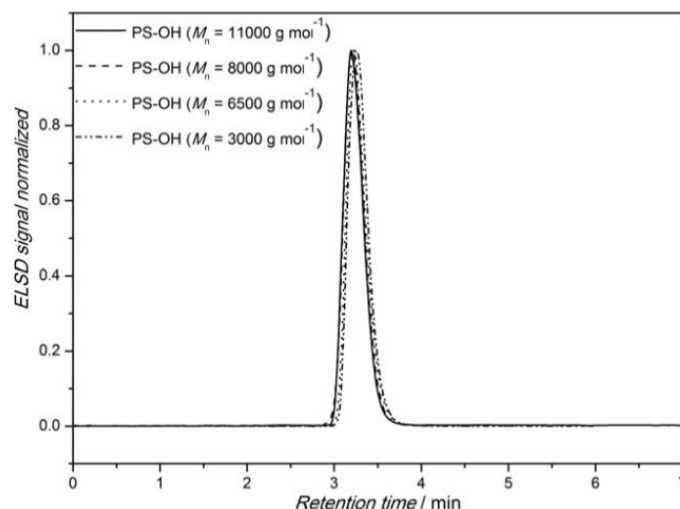
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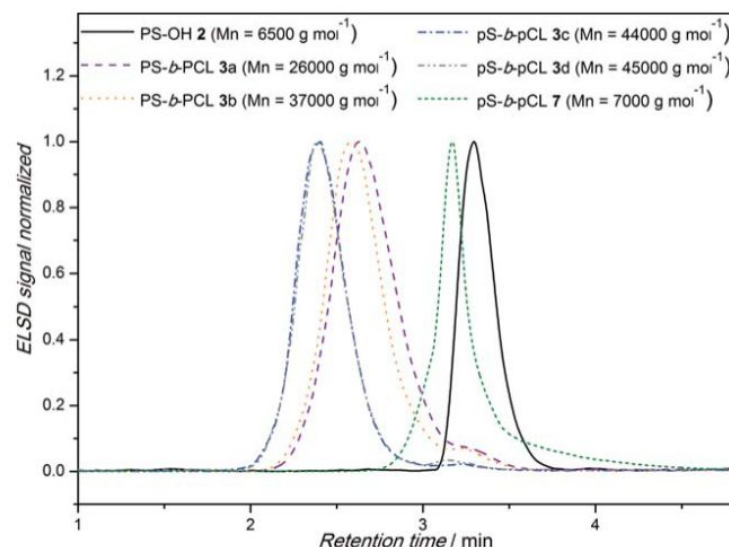


Coupling SEC to Other Characterization Methods

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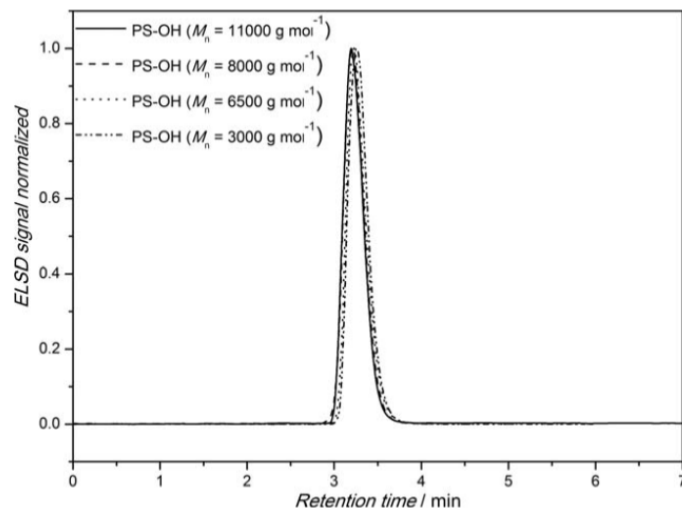


solvent mixture of THF:H₂O = 88.4 wt %:11.6 wt %
with addition of 0.1 vol % acetic acid

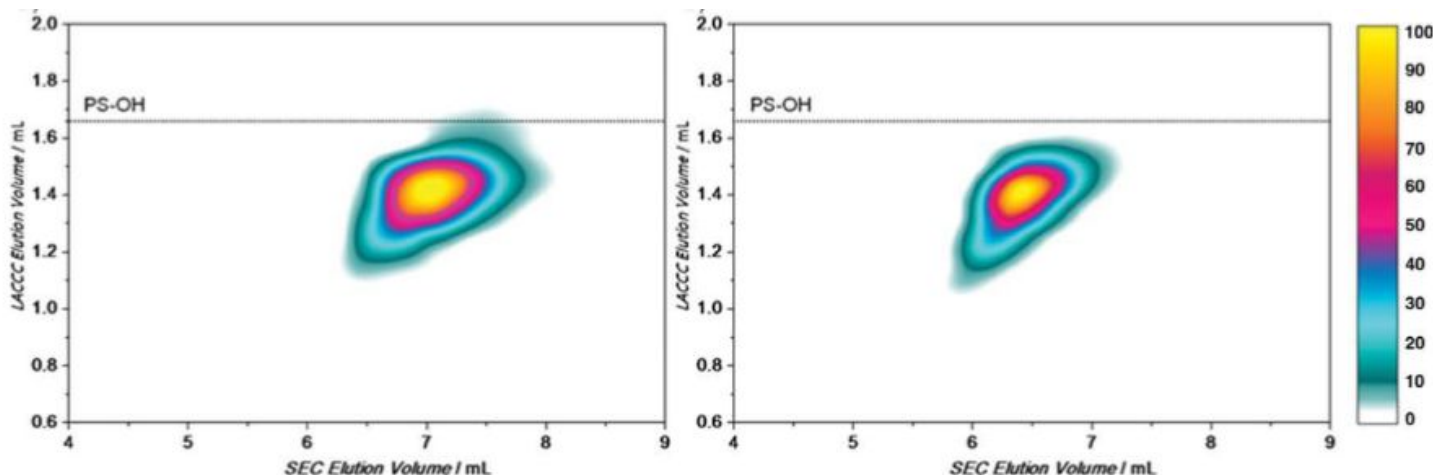
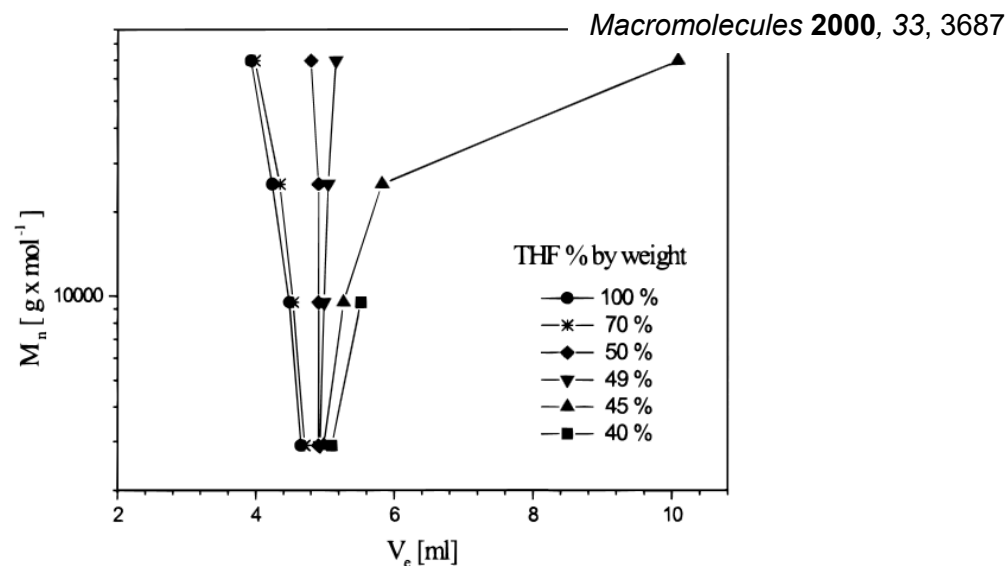


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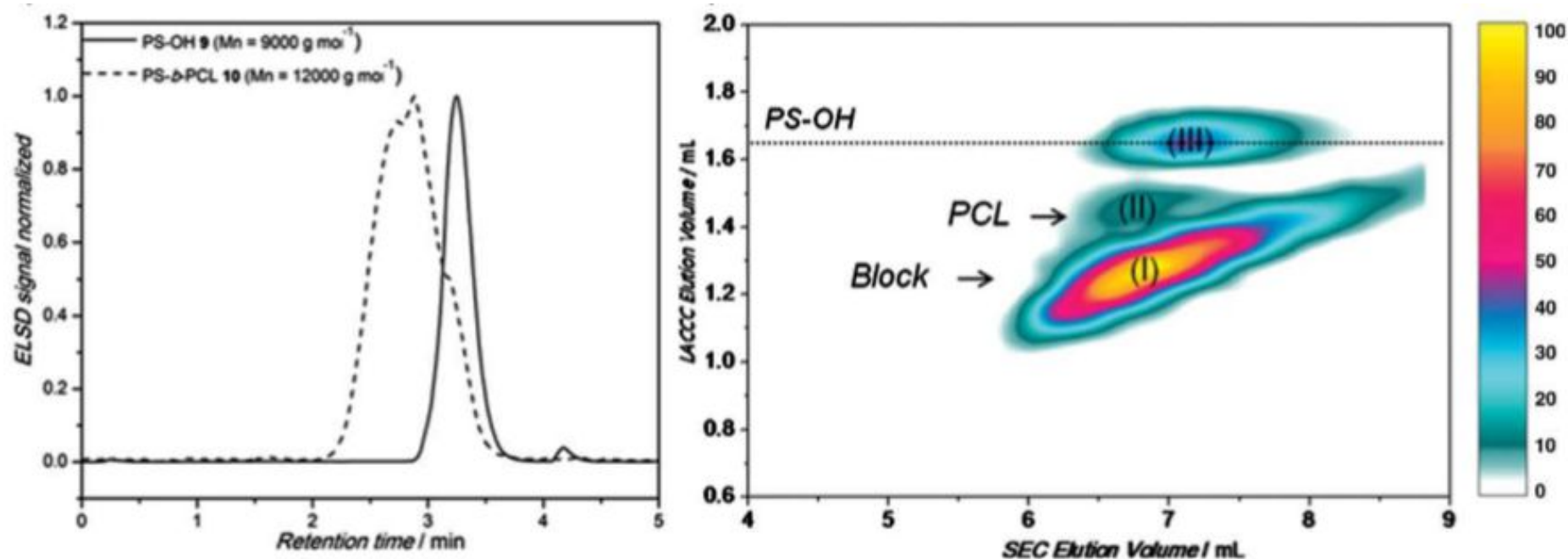
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Molecular Characterization of Block Copolymers

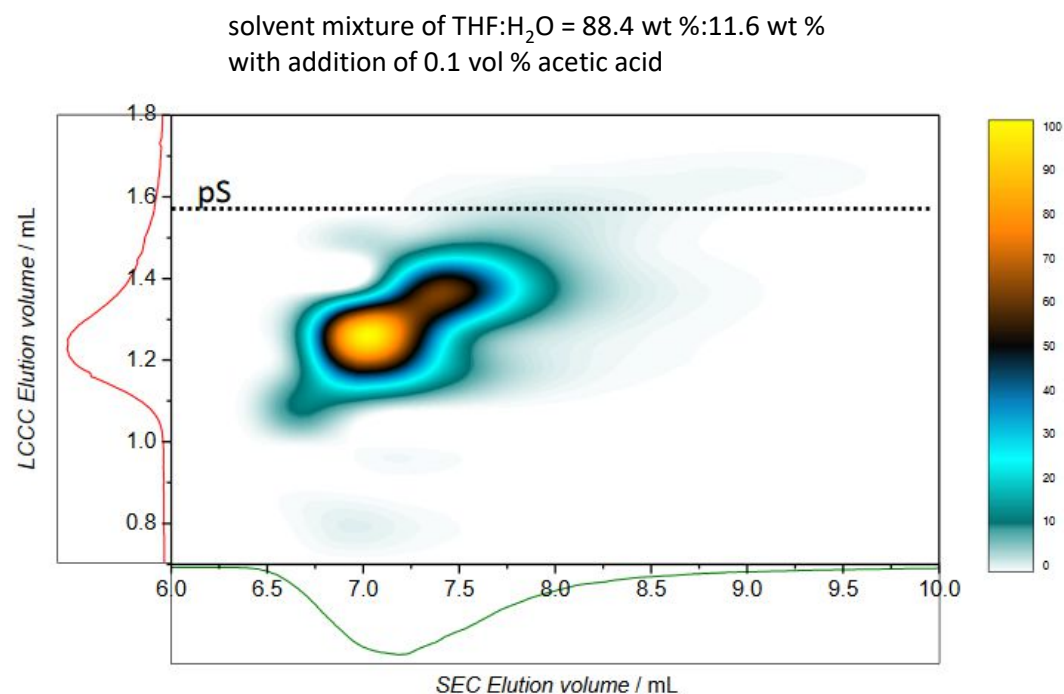
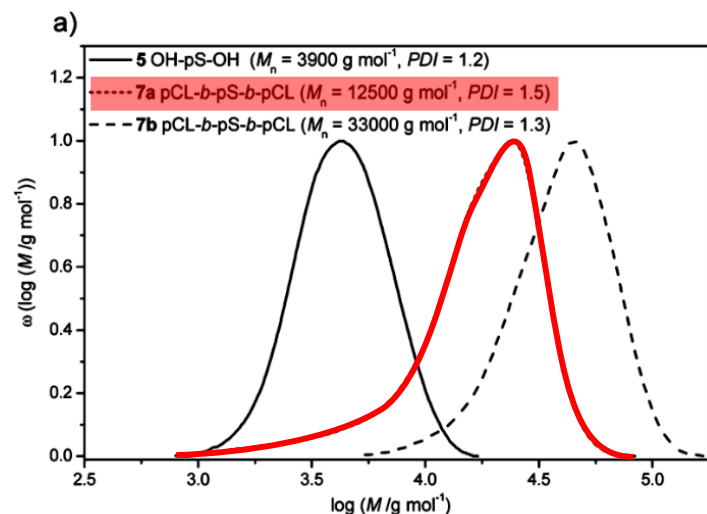
Coupling SEC to Other Characterization Methods

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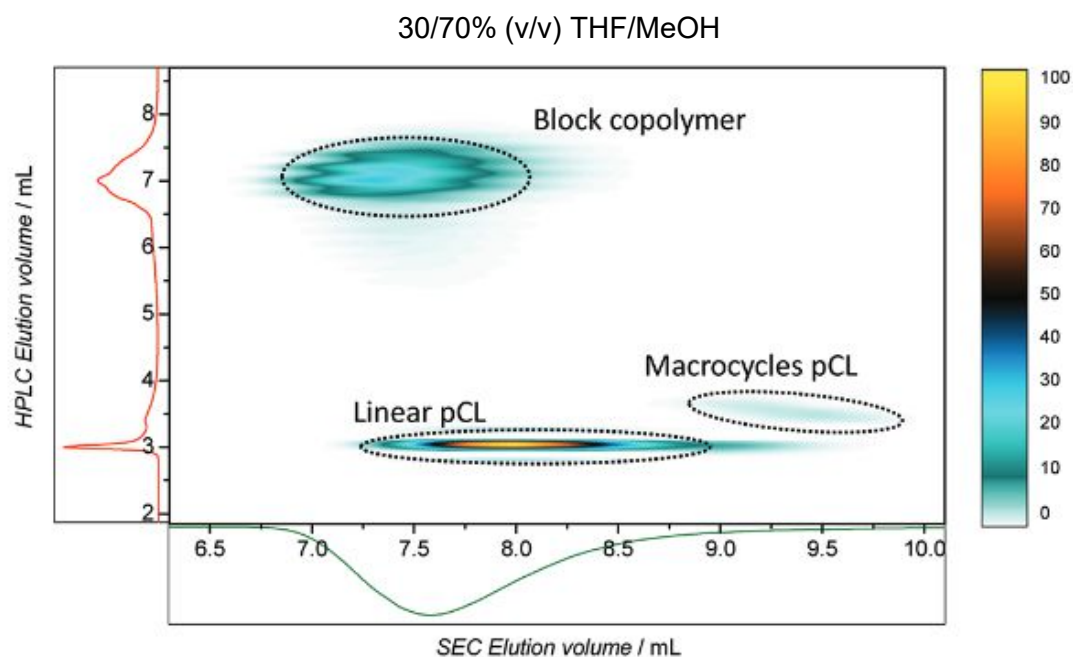
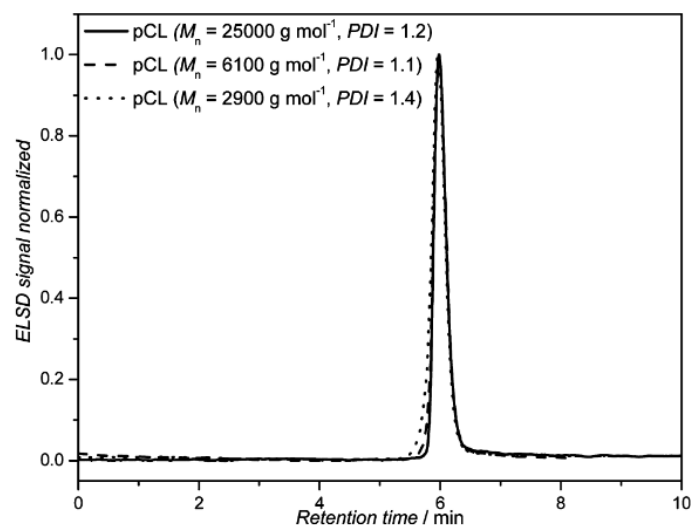
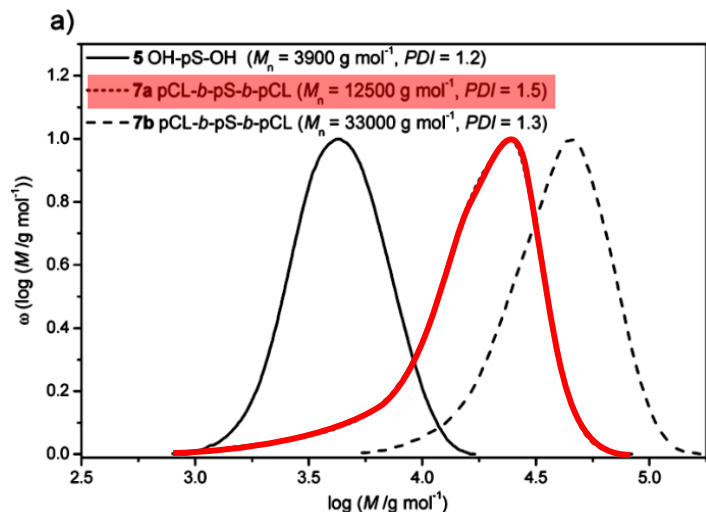
Coupling SEC to Other Characterization Methods

Liquid Adsorption Chromatography at Critical Conditions–SEC (LACCC–SEC)



Coupling SEC to Other Characterization Methods

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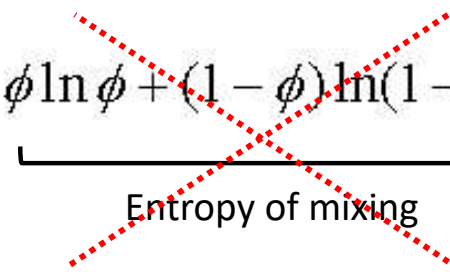
Macromolecules **2012**, 45, 87

(Purely) Synthetic Block Copolymers
- Solid-State Self-Assembly -
Why? How?

Why Do Polymers Segregate/Phase-Separate?

Thermodynamics: the system seeks to minimize free energy.

Flory-Huggins equation

$$\frac{\Delta G}{kT} = \underbrace{\phi \ln \phi + (1 - \phi) \ln(1 - \phi)}_{\text{Entropy of mixing}} + \underbrace{\chi \phi(1 - \phi)}_{\text{Enthalpy of mixing}}$$


The increase in entropy from mixing the two components together is drastically reduced in polymers: less mobility → less possible configurations.

Increasing the number of repeat units in the polymer (increasing molecular weight) decreases the entropic contribution to the free energy.

see
Flory–Huggins
theory lattice
(next slide)

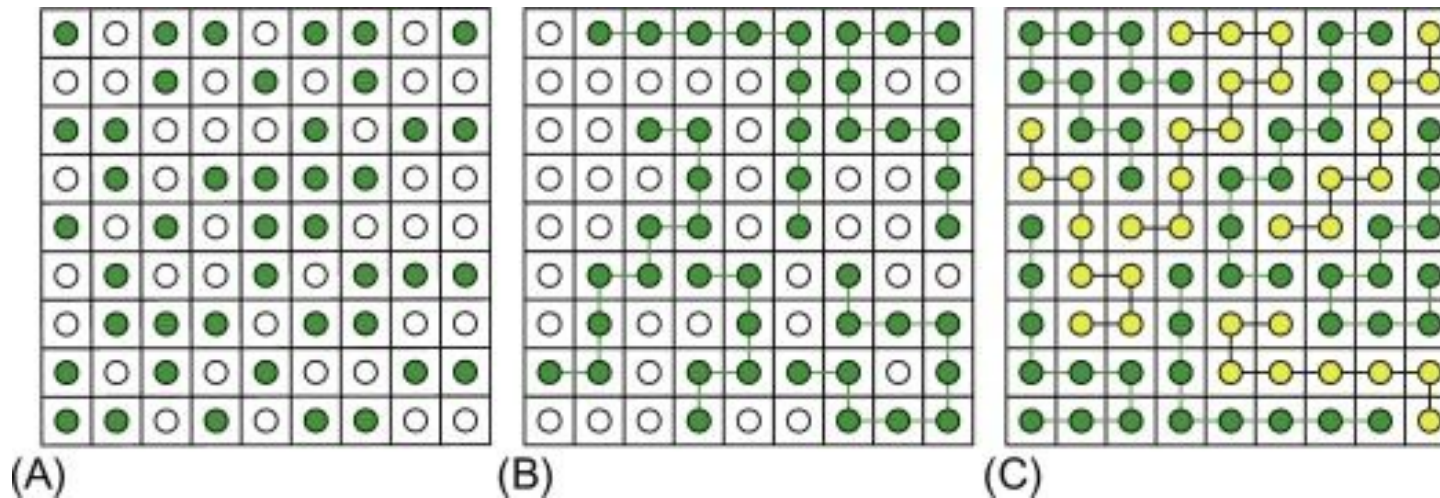
→ Enthalpy dominates and there are two minimums in the Gibbs free energy.

Why Do Polymers Segregate/Phase-Separate?

Thermodynamics: the system seeks to minimize free energy.

Flory-Huggins lattice theory

- no change of volume during mixing
- entropy of mixing is entirely given by the number of rearrangements during mixing
- enthalpy of mixing is caused by interactions of different segments after the dissolution of interactions of the same type of segments



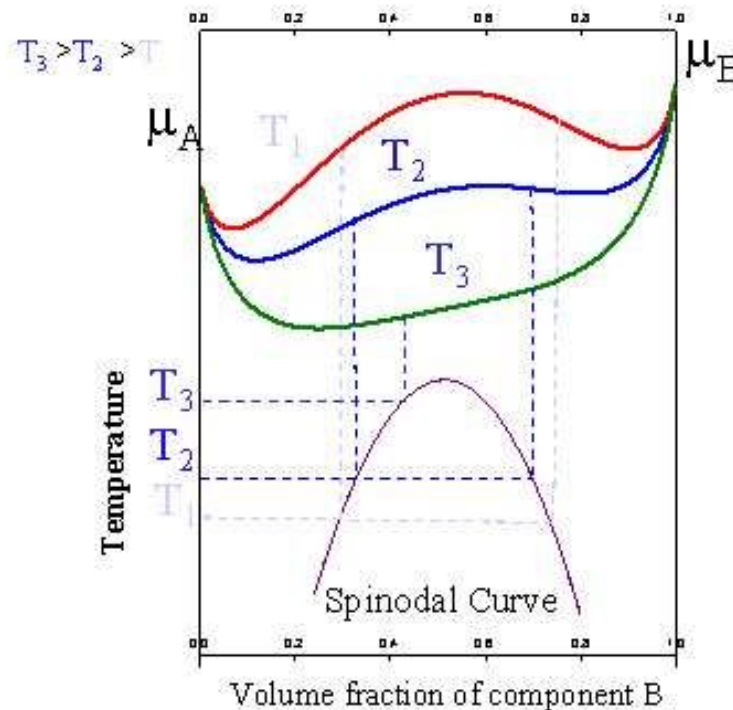
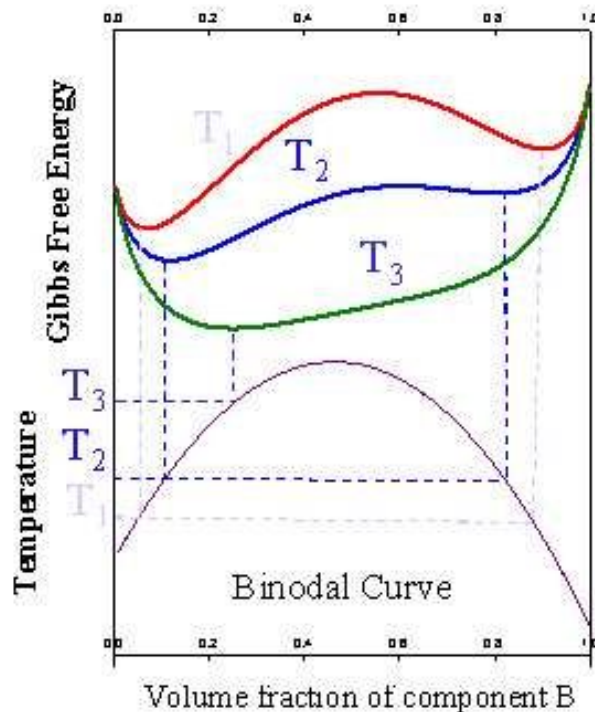
Theoretical Aspects of Solid-State Self-Assembly

Why Do Polymers Segregate/Phase-Separate?

Thermodynamics: the system seeks to minimize free energy.

Flory-Huggins equation

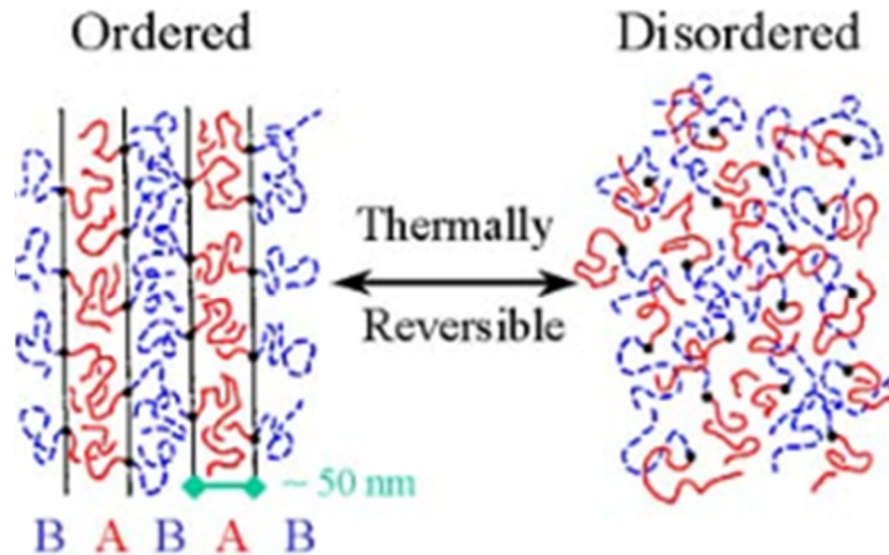
$$\frac{\Delta G}{kT} = \underbrace{\phi \ln \phi + (1 - \phi) \ln(1 - \phi)}_{\text{Entropy of mixing}} + \underbrace{\chi \phi(1 - \phi)}_{\text{Enthalpy of mixing}}$$



Spinodal decomposition
= rapid unmixing of a
mixture of liquids or solids

Block Copolymer Phase Separation

Phase Separation is Limited to the Nanoscale!



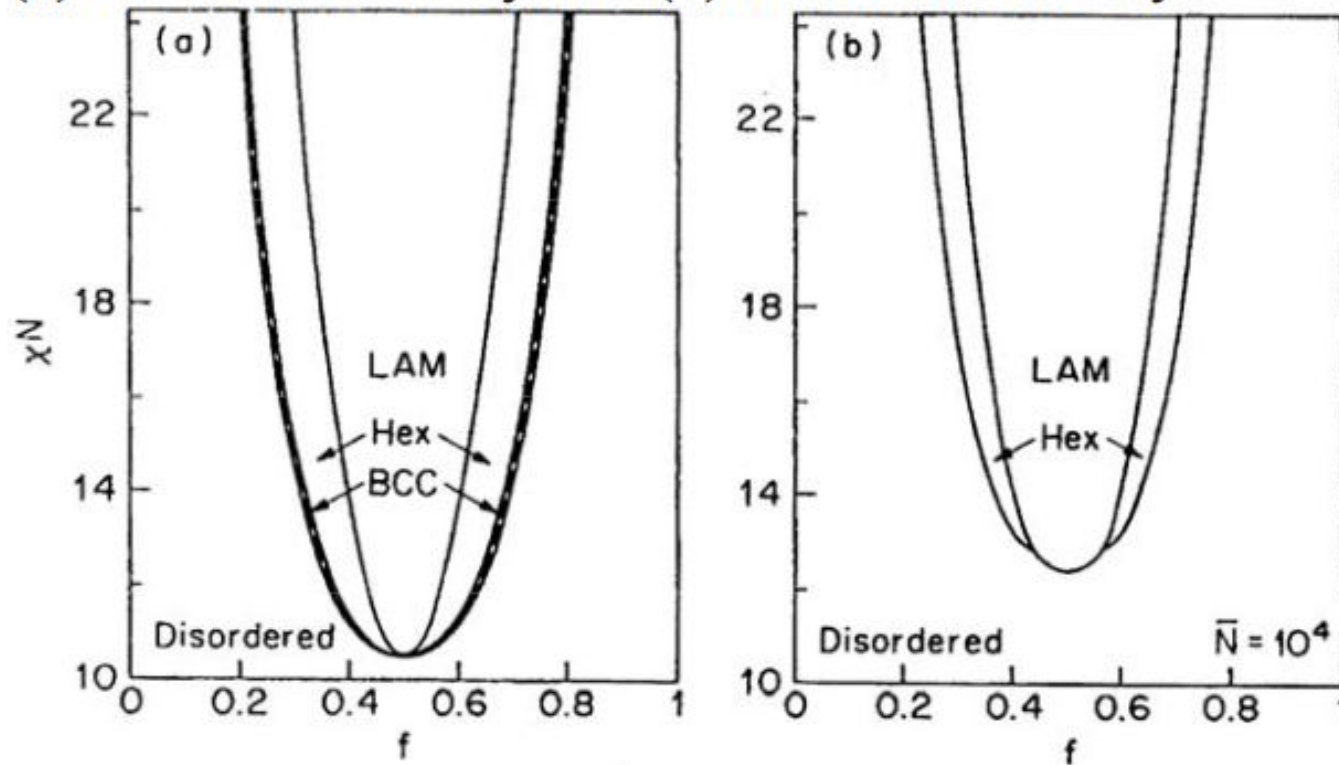
Important parameters

N degree of polymerization
 χ Flory-Huggins parameter
 f composition (block ratio)

Block Copolymer Phase Separation

Phase Separation is Limited to the Nanoscale!

(a) mean-field theory and (b) fluctuation theory



At a given
temperature

Important parameters

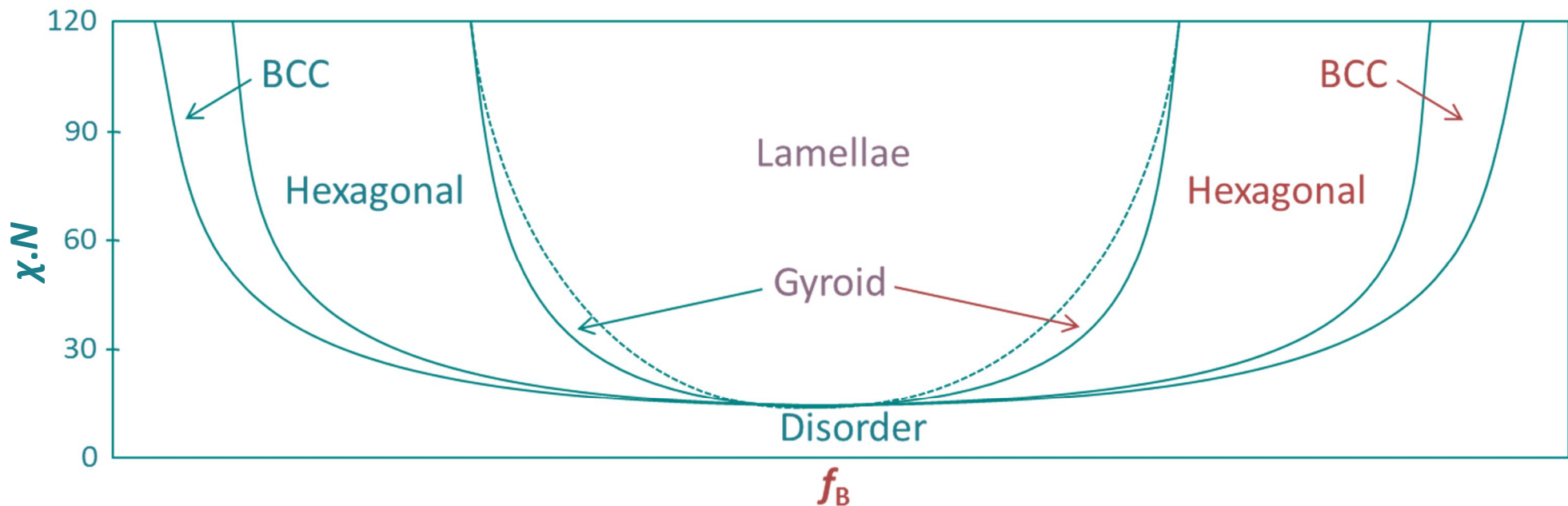
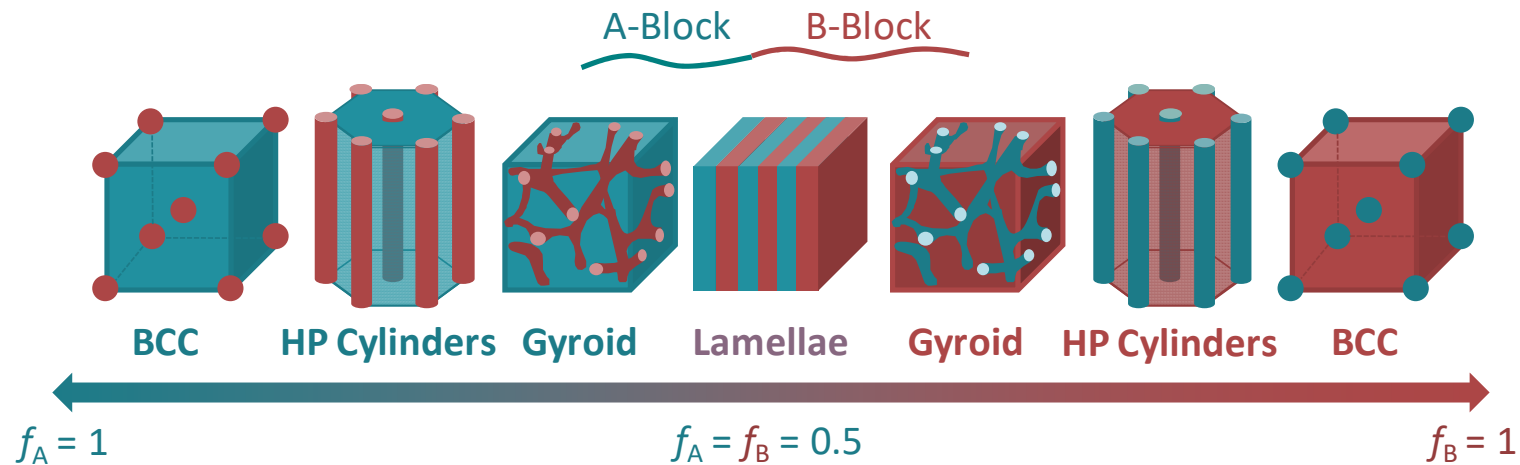
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Block Copolymer Phase Separation

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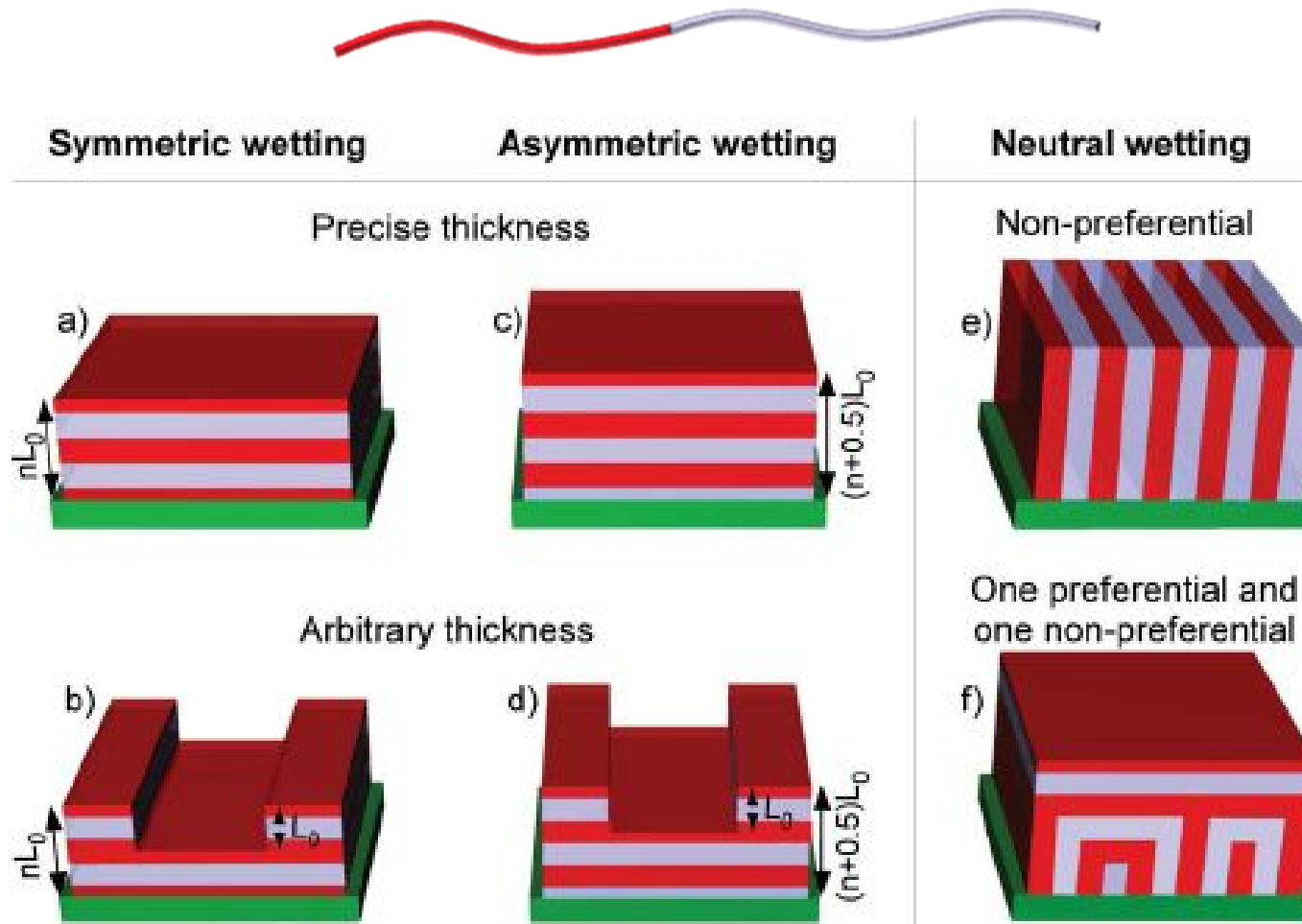


Macromolecules 1996, 29, 1091

Block Copolymer Phase Separation

Besides Block-Block Interactions... Interfaces

Influence on the orientation! Particularly important for thin films



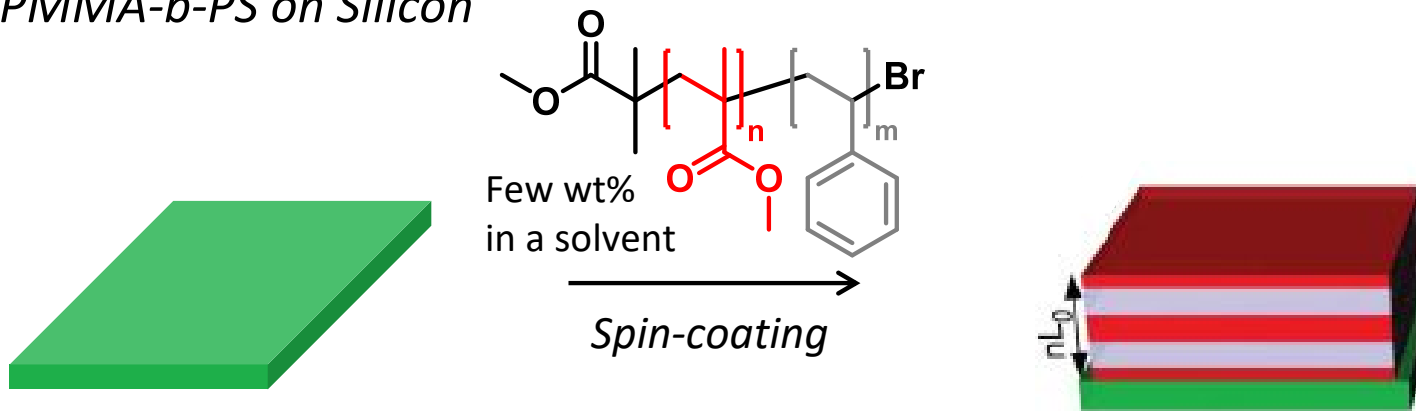
Particularly important for thin films

Soft Matter 2014, 10, 3867

Block Copolymer Phase Separation

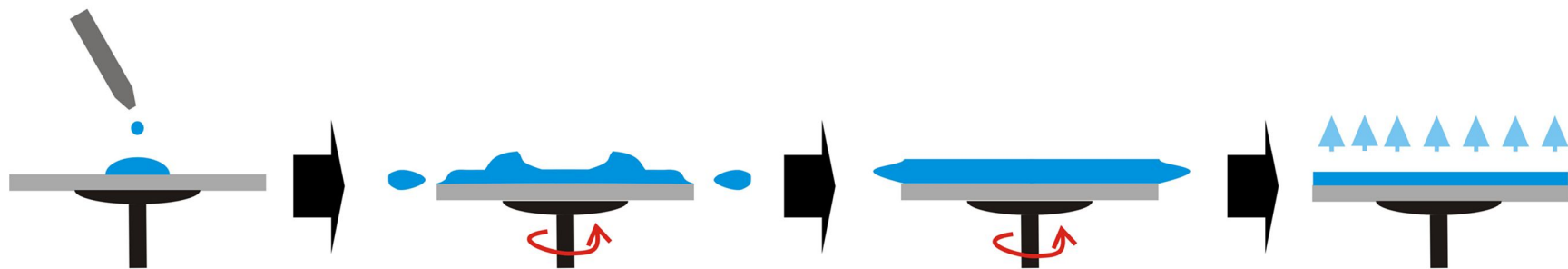
Besides Block-Block Interactions... Interfaces

Case of PMMA-*b*-PS on Silicon



Block Copolymer Phase Separation

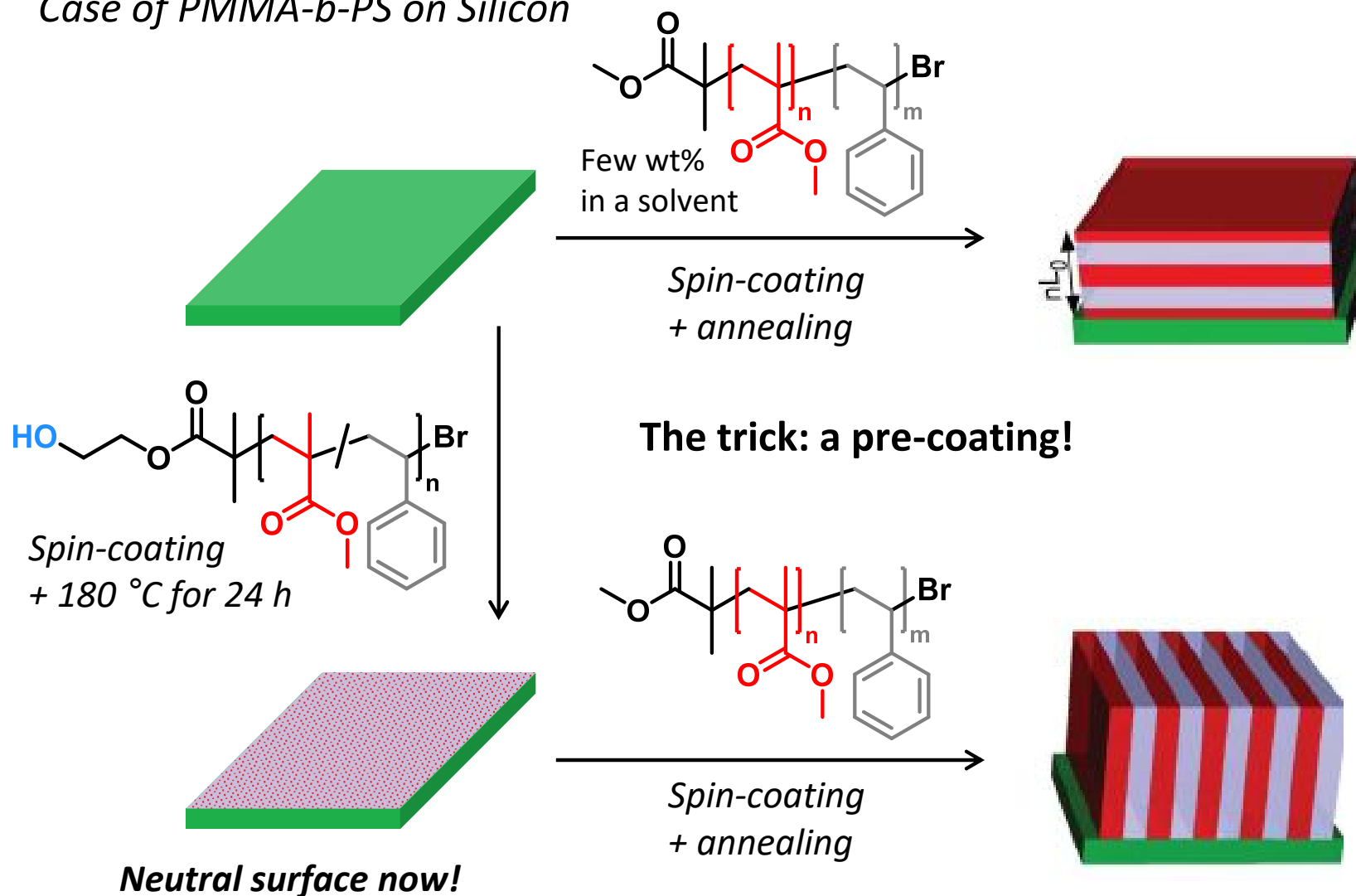
Thin Film Preparation by Spin-Coating



Block Copolymer Phase Separation

Besides Block-Block Interactions... Interfaces

Case of PMMA-*b*-PS on Silicon



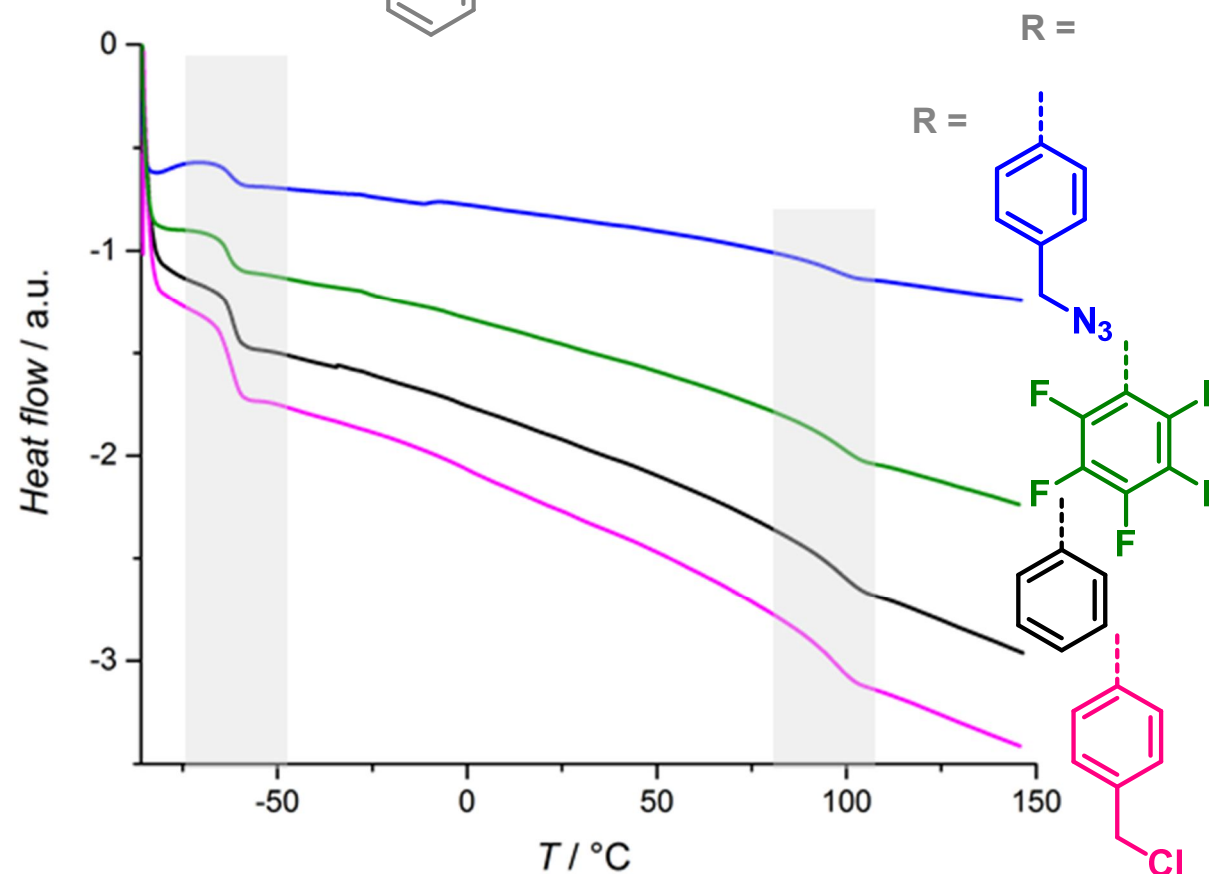
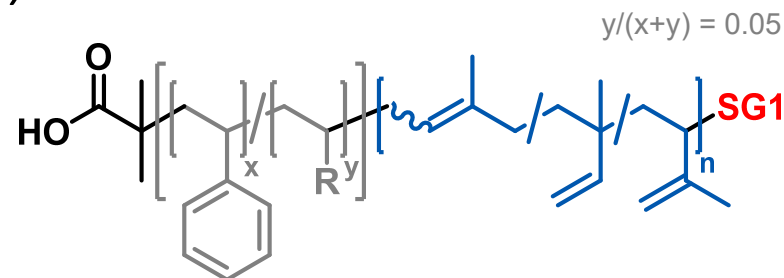
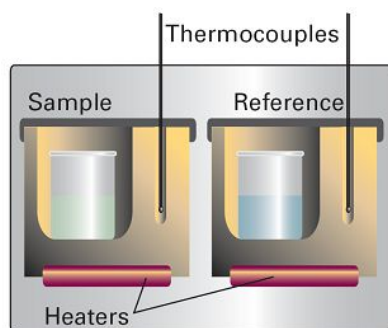
Alternatively, epoxides can be used for grafting or surface-initiated polymerization

(Purely) Synthetic Block Copolymers
- Solid-State Self-Assembly -
Characterization

Characterizing the Bulk Phase Separation

Differential Scanning Calorimetry

Observing the glass transition temperature(s)

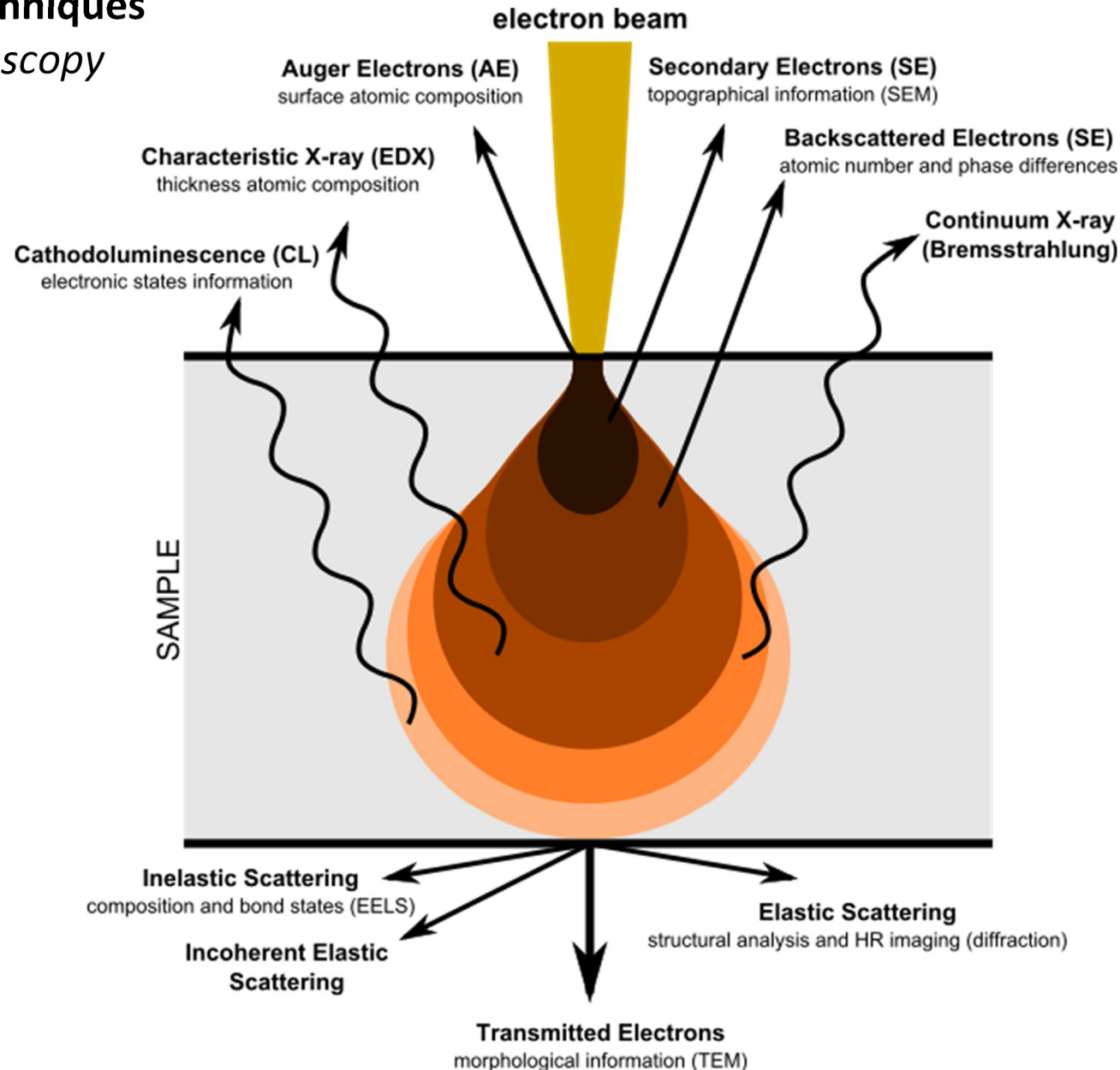


Delaittre group *Polym. Chem.* **2019**, 10, 1344

Characterizing the Bulk Phase Separation

“Imaging” Techniques

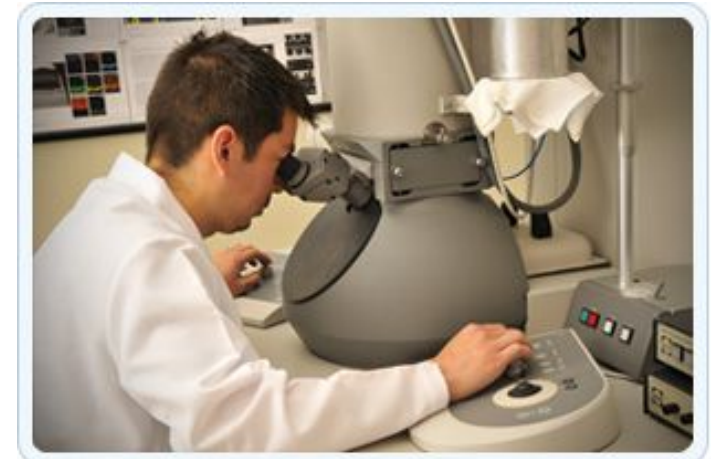
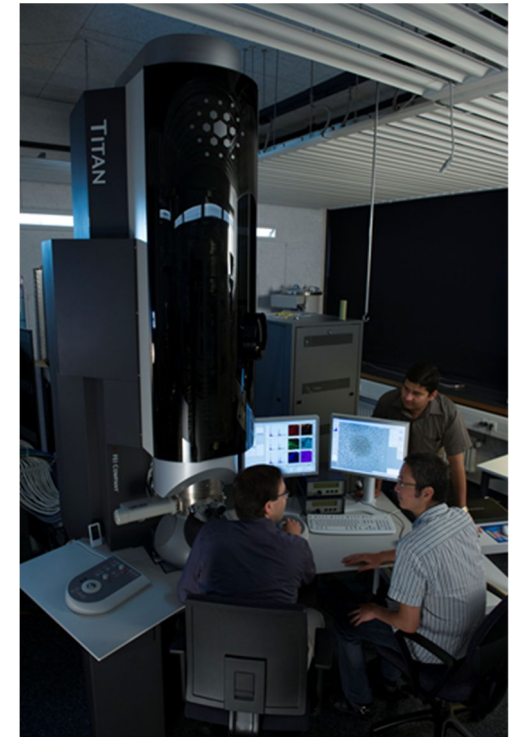
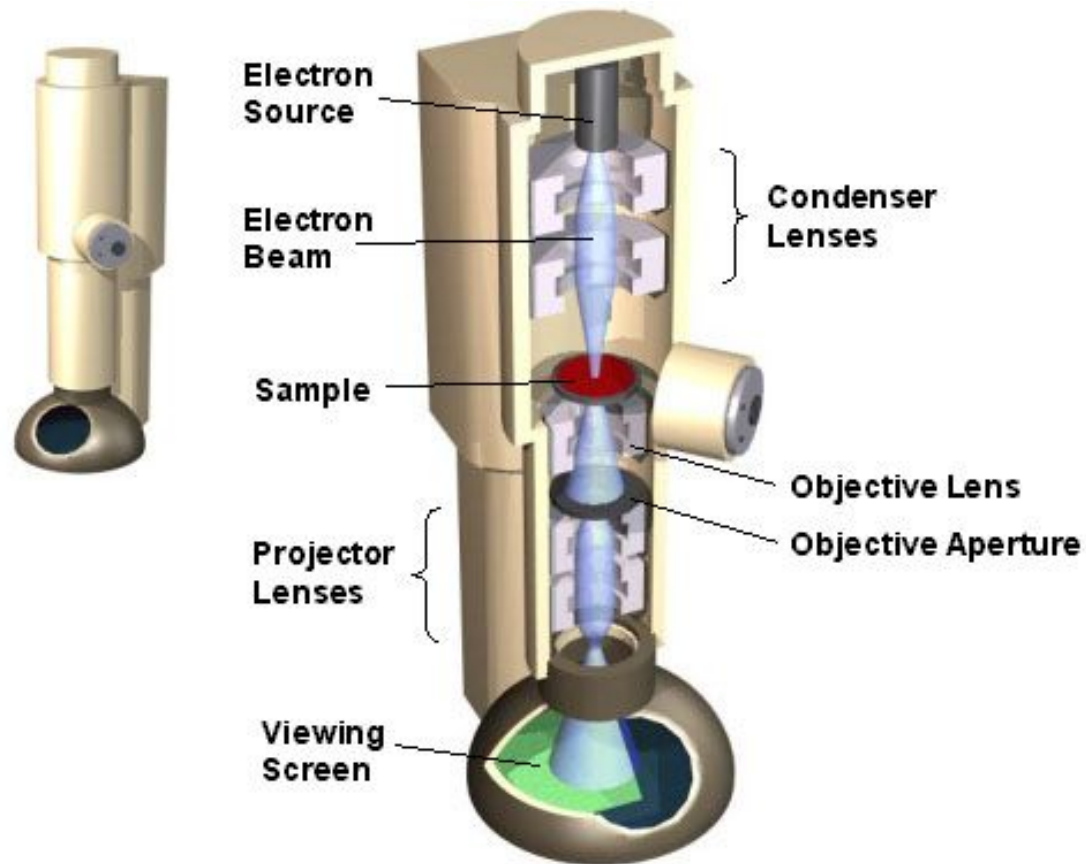
Electron Microscopy



Characterizing the Bulk Phase Separation

“Imaging” Techniques

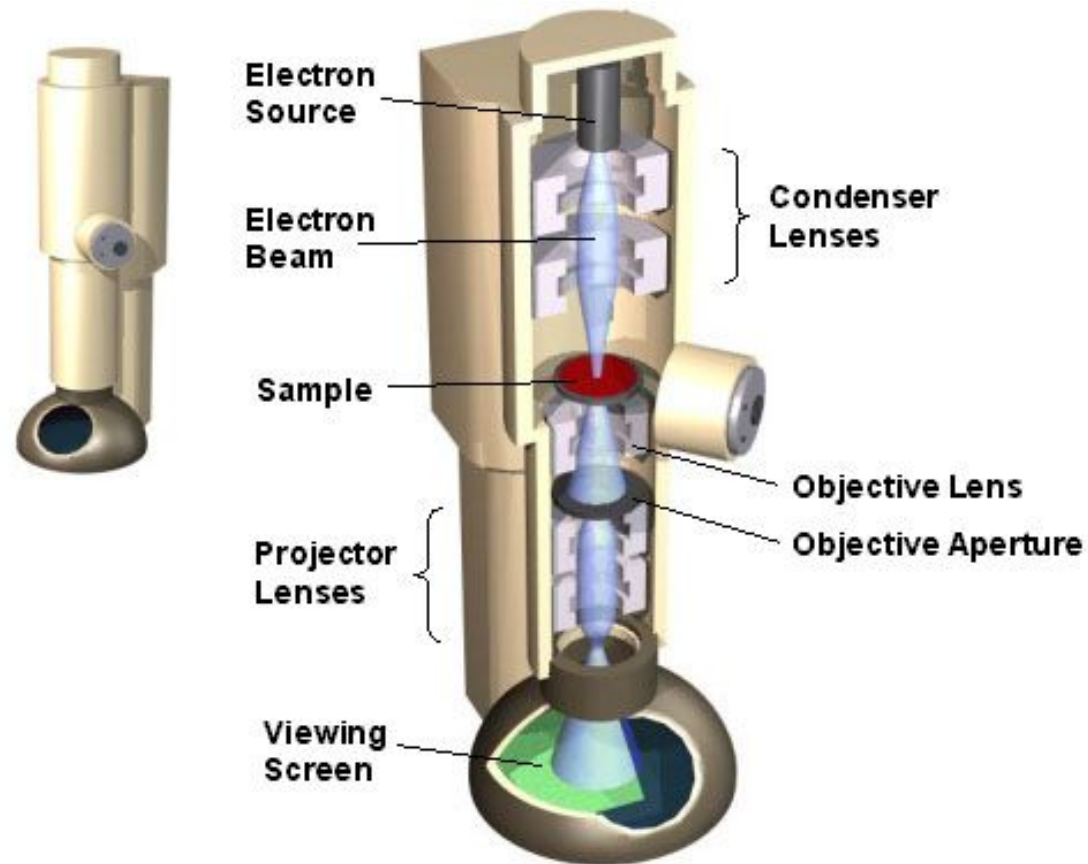
Transmission Electron Microscopy



Characterizing the Bulk Phase Separation

“Imaging” Techniques

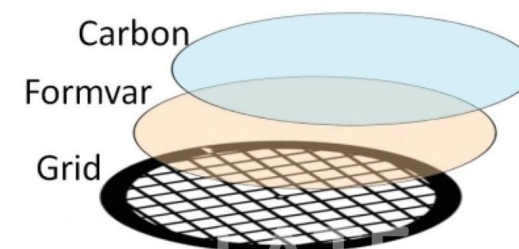
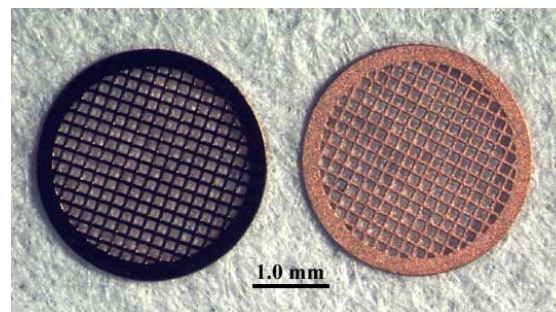
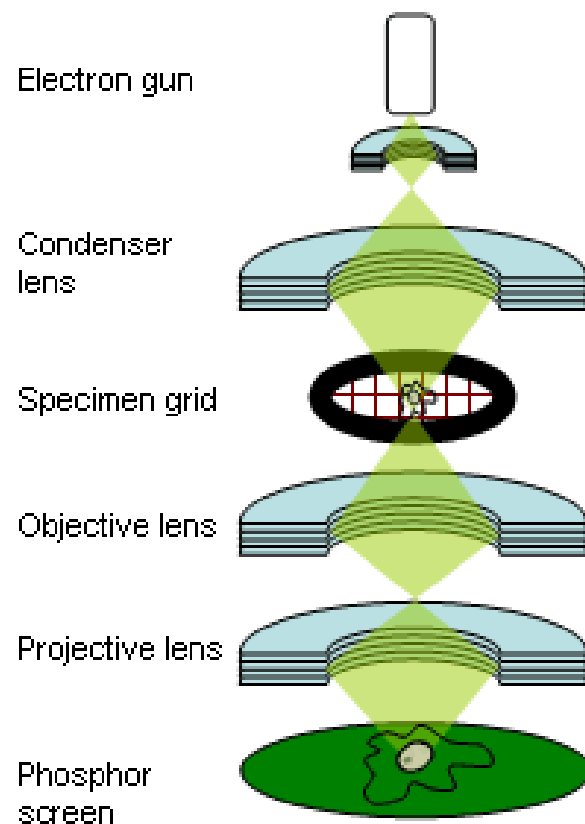
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Characterizing the Bulk Phase Separation

“Imaging” Techniques

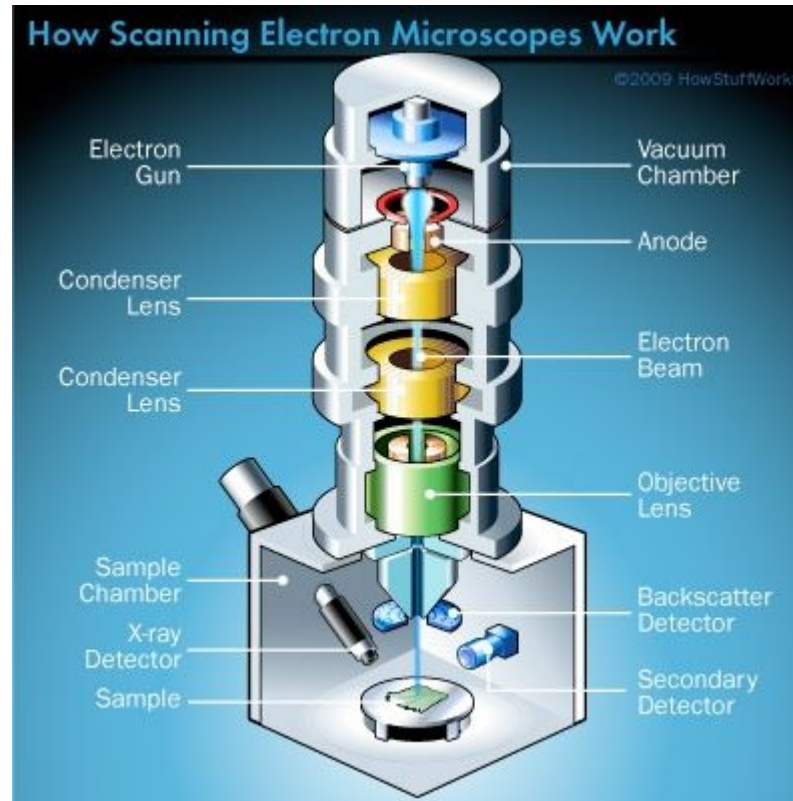
Transmission Electron Microscopy



Characterizing the Bulk Phase Separation

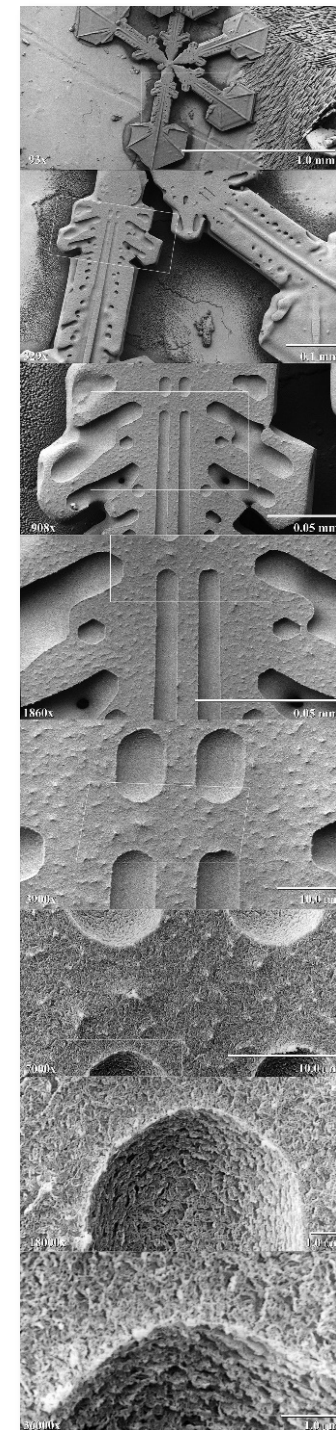
“Imaging” Techniques

Scanning Electron Microscopy



Samples are often sputtered with metals

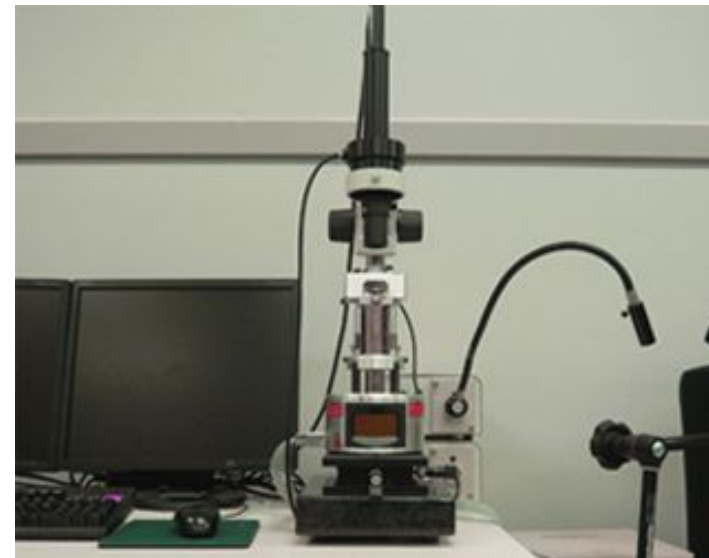
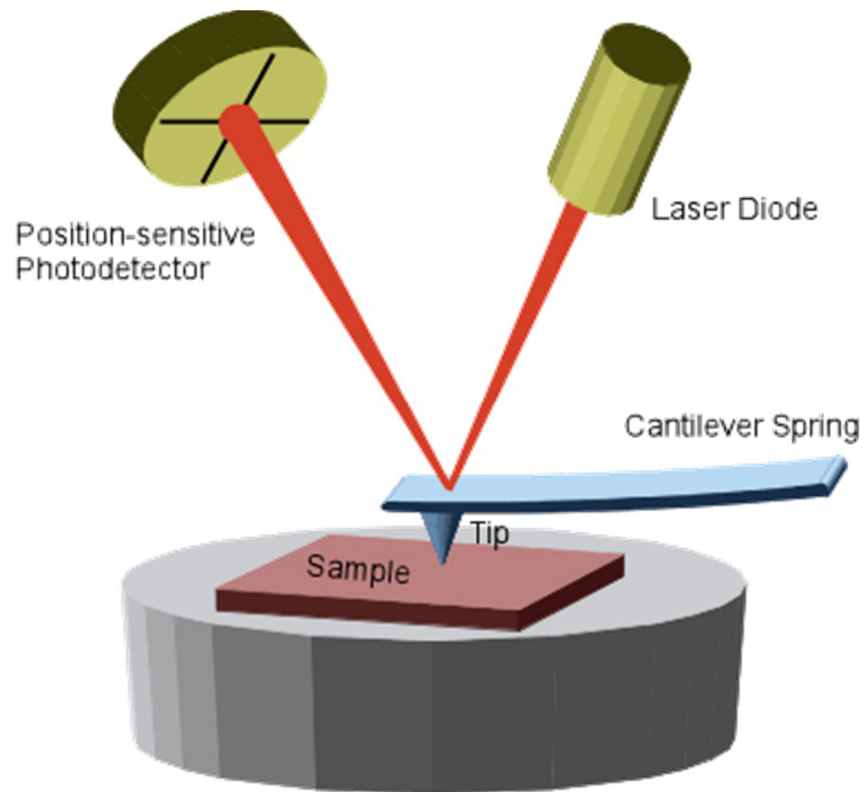
- Reduced microscope beam damage
- Reduced sample charging (increased conduction)
- Improved secondary electron emission
- Reduced beam penetration with improved edge resolution



Characterizing the Bulk Phase Separation

“Imaging” Techniques

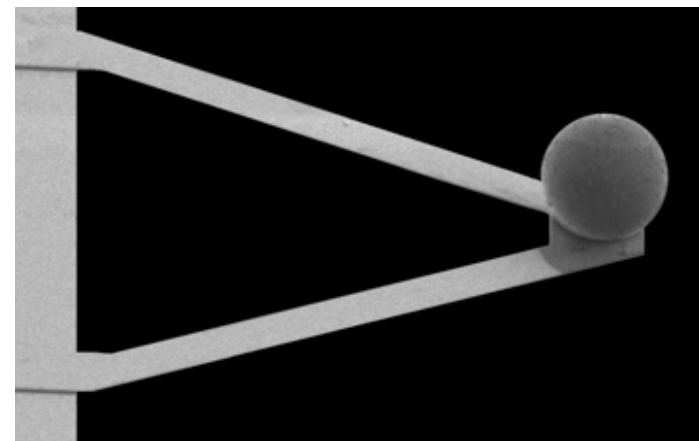
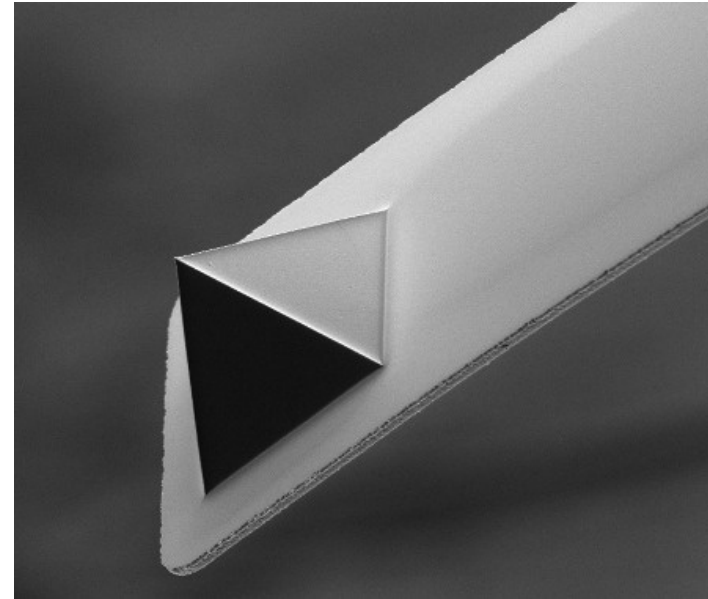
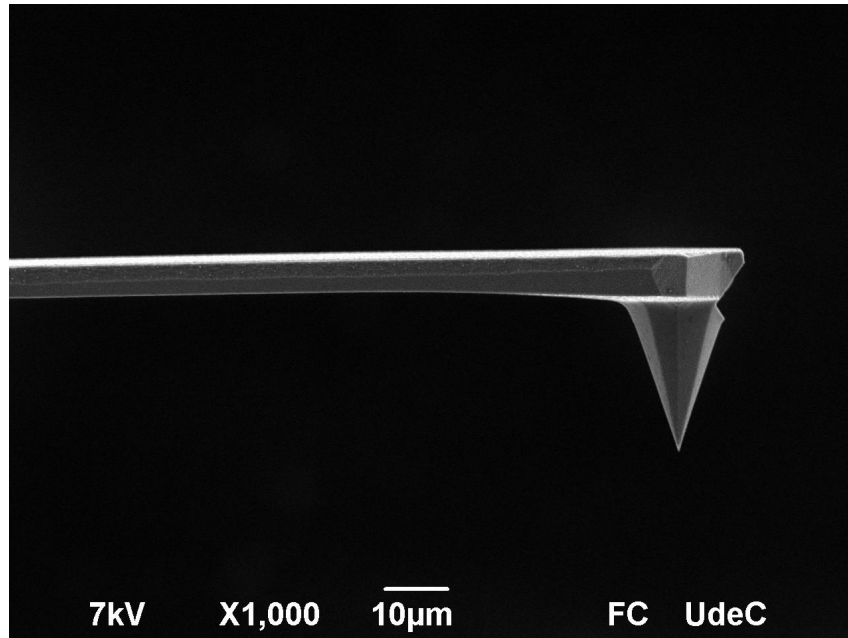
Atomic Force Microscopy (AFM)



Characterizing the Bulk Phase Separation

“Imaging” Techniques

Atomic Force Microscopy (AFM)

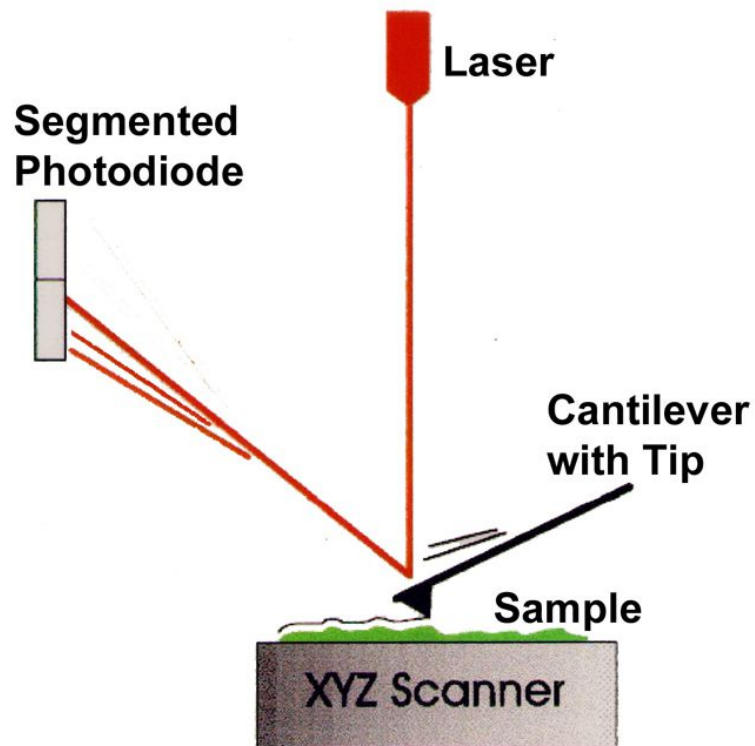


Characterizing the Bulk Phase Separation

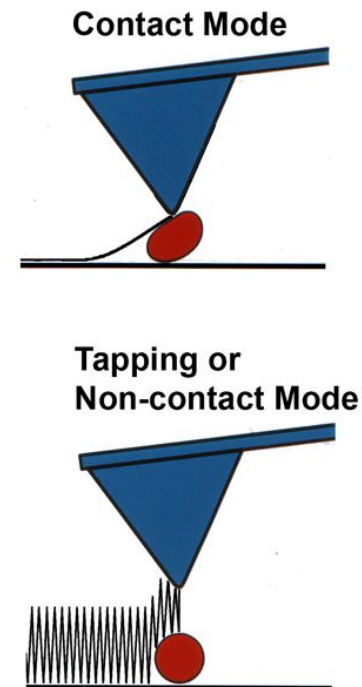
“Imaging” Techniques

Atomic Force Microscopy (AFM)

A Atomic Force Microscope



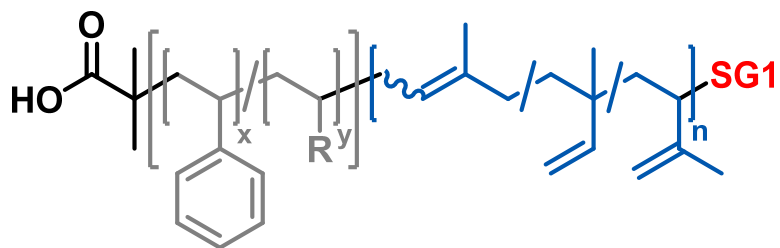
B AFM Imaging Modes



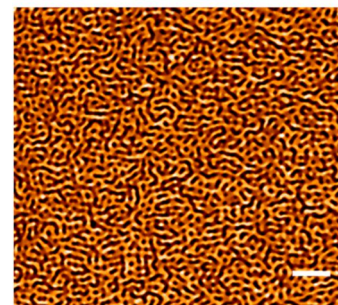
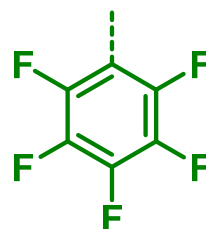
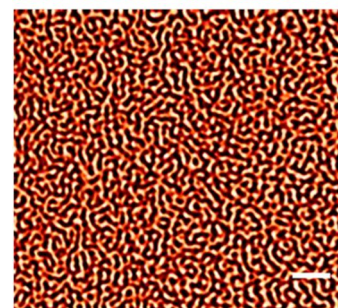
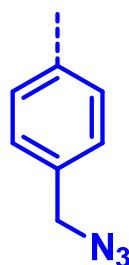
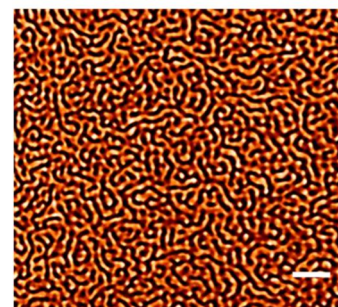
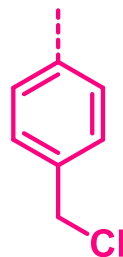
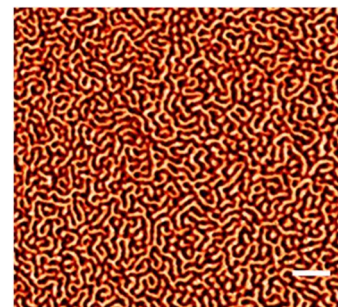
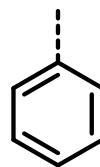
Characterizing the Bulk Phase Separation

“Imaging” Techniques

Atomic Force Microscopy (AFM)



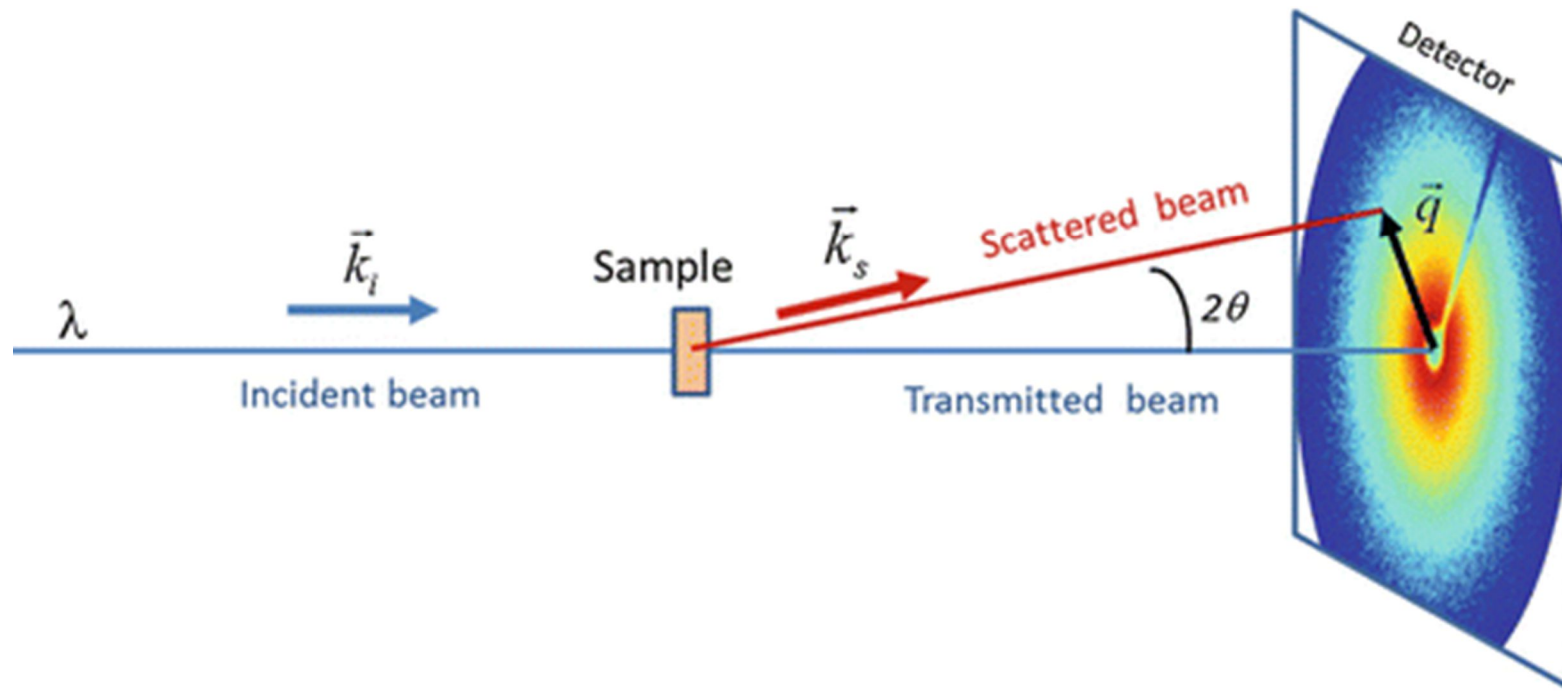
R =



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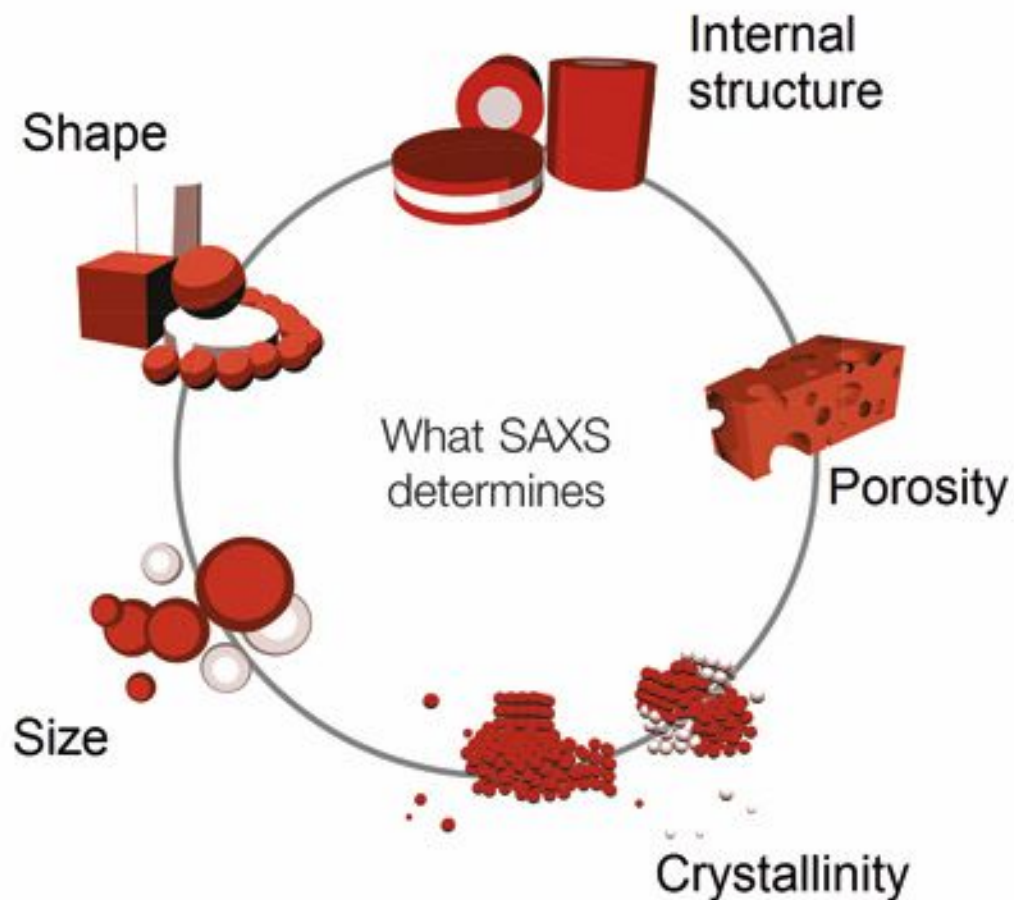
Characterizing the Bulk Phase Separation

Small-Angle X-Ray Scattering (SAXS)



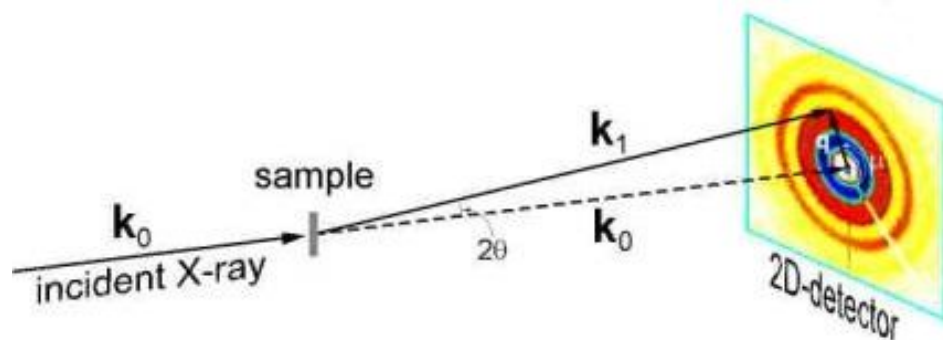
Characterizing the Bulk Phase Separation

Small-Angle X-Ray Scattering (SAXS)

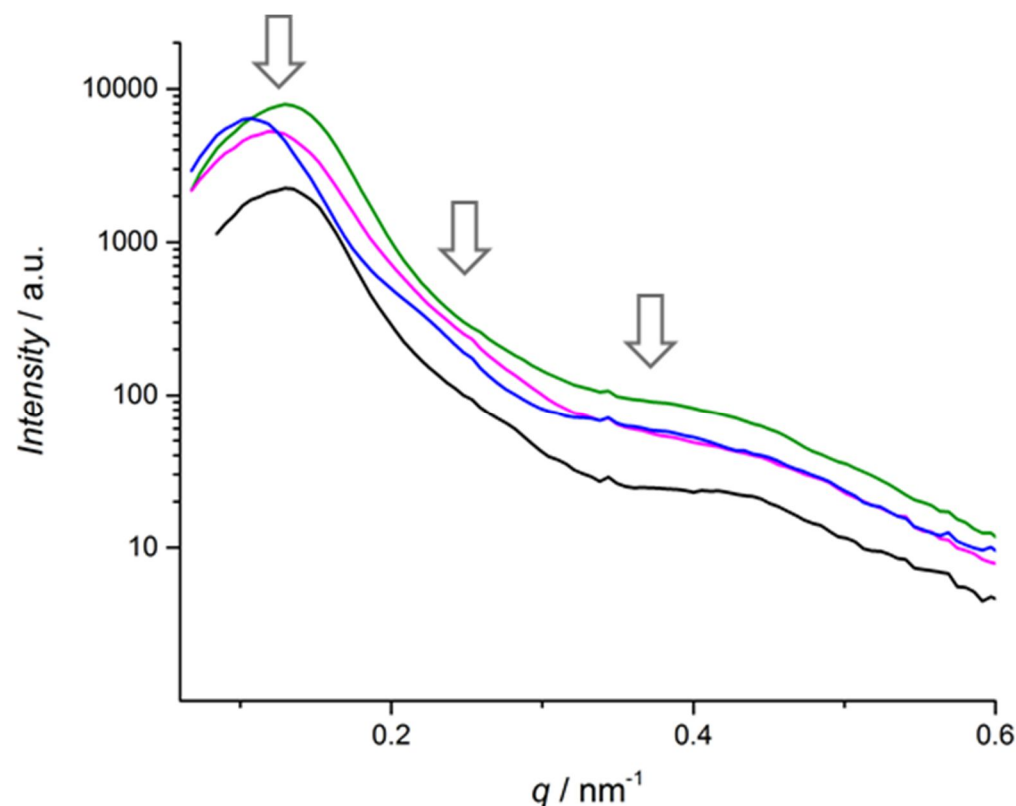
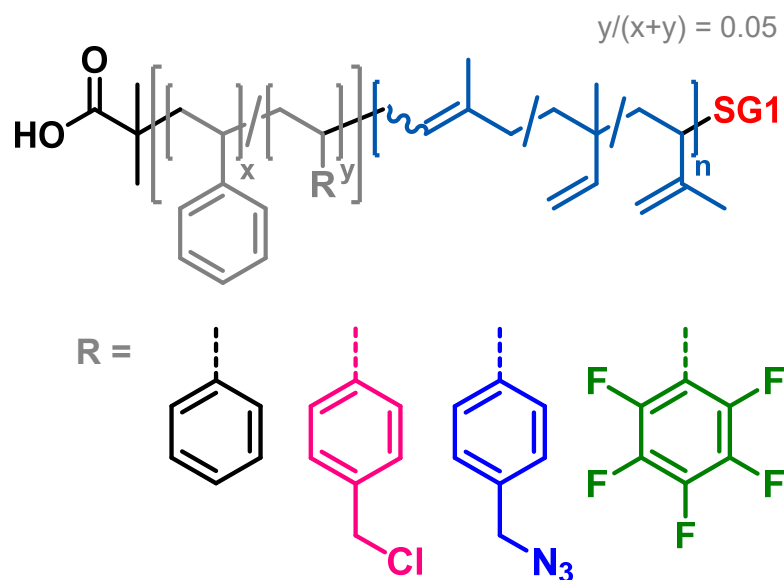


Characterizing the Bulk Phase Separation

Small-Angle X-Ray Scattering (SAXS)



scattering vector $q = 4\pi \sin(\vartheta)/\lambda$

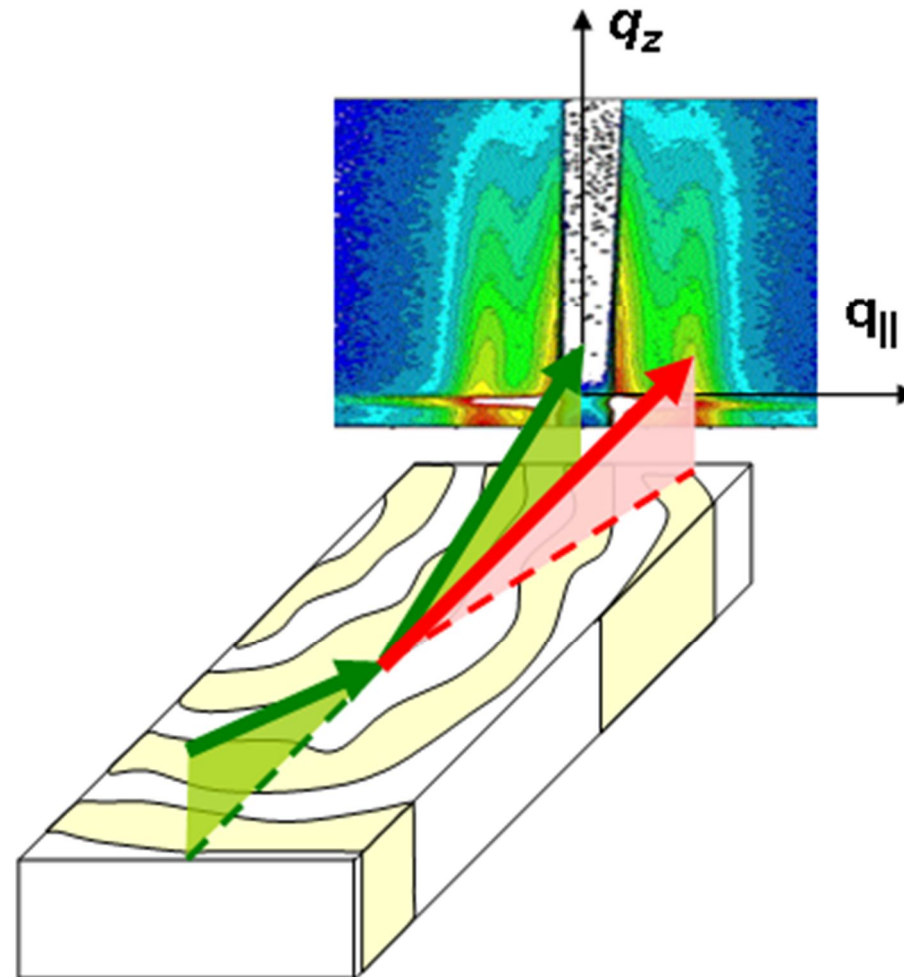


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Characterizing the Bulk Phase Separation

Small-Angle X-Ray Scattering (SAXS)

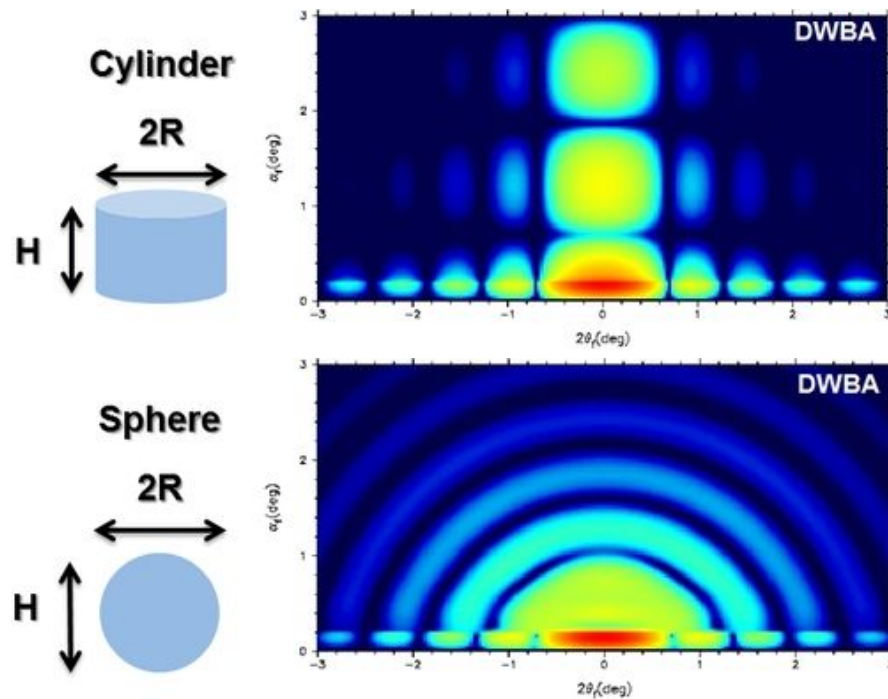
In reflection mode: Grazing Incidence SAXS (GISAXS)



Characterizing the Bulk Phase Separation

Small-Angle X-Ray Scattering (SAXS)

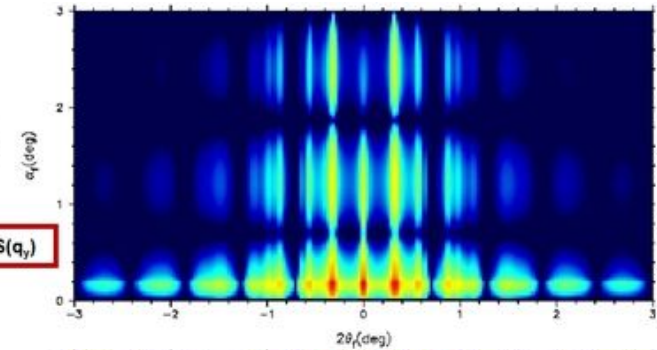
In reflection mode: Grazing Incidence SAXS (GISAXS)



Cylinders

**2D-
hexagonal
lattice**

$$I(q_y, q_z) \sim |F(q_y, q_z)|^2 \times S(q_y)$$



Spheres

**2D-
hexagonal
lattice**

$$I(q_y, q_z) \sim |F(q_y, q_z)|^2 \times S(q_y)$$

